

A Quadratic Tower-Diagonal Lower Bound for Approximate-Private PAC Learning

Abstract

We prove a nonasymptotic sample lower bound for central approximate-private, distribution-free realizable PAC learning of tagged products of finite threshold classes. At fixed constant accuracy, confidence, and ε , with δ in the stated candidate regime, the bound is $\Omega(k \log_2^* N)$. On the tower diagonal $k = r$, $N = \text{Tow}_2(r)$, it becomes $\Omega(r^2) = \Omega(\text{VC} \log_2^* \text{LD})$ for unrestricted randomized improper learners. The result separates this family from the generic additive scale, but it does not establish the stronger exponential or log-cardinality lower bounds posed by the general open problem.

1 Theoretical Setup

1.1 Tagged threshold products

For a positive integer q , write $[q] = \{1, \dots, q\}$. Fix integers $k \geq 2$ and $N \geq 2$, and define

$$X_{k,N} = [k] \times [N], \quad \tau_t(x) = \mathbf{1}\{x \geq t\} \quad (t \in [N+1], x \in [N]).$$

The tagged threshold-product class and the unrestricted improper output space are

$$C_{k,N} = \{c_t : t \in [N+1]^k\}, \quad c_t(i, x) = \tau_{t_i}(x), \quad \mathcal{H}_{k,N} = \{0, 1\}^{X_{k,N}}.$$

No properness, representation, or computational restriction is imposed.

For a distribution P on $X_{k,N}$, a target $c_t \in C_{k,N}$, and $h \in \mathcal{H}_{k,N}$, let

$$R_P(h, c_t) = \Pr_{Z \sim P} \{h(Z) \neq c_t(Z)\}.$$

Write P^{c_t} for the realizable labeled law of $(Z, c_t(Z))$ with $Z \sim P$. Thus $S \sim (P^{c_t})^n$ always denotes exactly n independent labeled examples.

A candidate learner is an arbitrary randomized map

$$A : (X_{k,N} \times \{0, 1\})^n \longrightarrow \mathcal{H}_{k,N}.$$

Two ordered size- n labeled datasets are adjacent, denoted $S \simeq S'$, when they differ in at most one row by replacement. Every unadorned log is natural, and

$$\log_2^* u = \min\{j \in \mathbb{Z}_{\geq 0} : \log_2^{(j)} u \leq 1\}, \quad m_{n,k} = \max\left\{8, \left\lceil \frac{4n}{k} \right\rceil\right\}.$$

We use the tower convention

$$\text{Tow}_2(1) = 2, \quad \text{Tow}_2(r+1) = 2^{\text{Tow}_2(r)}.$$

1.2 Assumptions

The constants in the following assumptions are fixed before the candidate parameters and learner are chosen.

Assumption 1 (Primitive candidate regime). *For fixed $c_\delta, \varepsilon_0 > 0$ and $N_0 \in \mathbb{Z}_{\geq 2}$, the candidate parameters satisfy*

$$k \geq 2, \quad N \geq N_0, \quad n \in \mathbb{Z}_{\geq 1}, \quad 0 < \varepsilon \leq \varepsilon_0,$$

and, with $M = m_{n,k}$,

$$0 < \delta \leq \min \left\{ \frac{1}{n \log(n+1)}, \frac{c_\delta}{M^2 \log(M+1)} \right\}.$$

Assumption 2 (Central approximate differential privacy). *For every $S \simeq S'$ in $(X_{k,N} \times \{0,1\})^n$ and every event $E \subseteq \mathcal{H}_{k,N}$,*

$$\Pr\{A(S) \in E\} \leq e^\varepsilon \Pr\{A(S') \in E\} + \delta,$$

where the probabilities are over only the internal randomness of A .

Assumption 3 (Distribution-free realizable PAC guarantee). *For fixed $\alpha_0, \beta_0 \in (0, 1/2)$, every $\mathbf{t} \in [N+1]^k$ and every distribution P on $X_{k,N}$ satisfy*

$$\Pr_{\substack{S \sim (P^{\mathbf{c}_t})^n \\ A}} \{R_P(A(S), \mathbf{c}_t) \leq \alpha_0\} \geq 1 - \beta_0.$$

The probability includes the exact i.i.d. sample and the learner's internal randomness.

1.3 Objective

We seek a deterministic, pointwise-in- n lower-bound implication. Its probabilistic premise is the fixed-size PAC guarantee above; its conclusion does not average over candidate sample sizes, targets, or distributions. The loss throughout is population zero-one risk.

2 Preliminaries

Let $\Delta([N])$ be the simplex of probability laws on $[N]$, and set

$$\mathcal{I}_N = [N+1] \times \Delta([N]).$$

For $(t, Q) \in \mathcal{I}_N$, let Q^{τ_t} be the law of $(X, \tau_t(X))$ for $X \sim Q$, and define the one-block population risk

$$R_Q(g, \tau_t) = \Pr_{X \sim Q} \{g(X) \neq \tau_t(X)\}, \quad g \in \{0,1\}^{[N]}.$$

For a deterministic vector $((t_i, Q_i))_{i=1}^k \in \mathcal{I}_N^k$, write

$$P_Q(i, x) = \frac{1}{k} Q_i(x), \quad \mathbf{c}_t(i, x) = \tau_{t_i}(x).$$

This notation is used for fixed product instances; when the vector itself is random, every i.i.d. assertion is conditional on its realized value.

For $h \in \mathcal{H}_{k,N}$ and $j \in [k]$, define the block restriction

$$(D_j h)(x) = h(j, x), \quad x \in [N],$$

and, for deterministic $\mathbf{t}, \mathbf{Q}$, the block error

$$e_j(h; \mathbf{t}, \mathbf{Q}) = \sum_{x=1}^N Q_j(x) \mathbf{1}\{h(j, x) \neq \tau_{t_j}(x)\}.$$

The finite spaces used in the simulator are

$$\mathbf{Z}_M = ([N] \times \{0, 1\})^M, \quad \mathbf{S}_n = (X_{k,N} \times \{0, 1\})^n, \quad \mathbf{G}_N = \{0, 1\}^{[N]}.$$

A randomized map between any of these finite spaces is identified with its stochastic kernel. All differential-privacy comparisons use ordered replacement-adjacent input pairs and arbitrary output events.

Finally, $\text{VC}(C)$ and $\text{LD}(C)$ denote the VC and Littlestone dimensions of a binary class C . The proof below uses complete binary instance trees for the latter and gives the needed tree and mistake-budget arguments directly.

3 Main Theorem

Theorem 3.1 (Tagged threshold-product lower bound). *There exist absolute constants*

$$a > 0, \quad c_\delta > 0, \quad N_0 \in \mathbb{Z}_{\geq 2},$$

for which one may fix

$$\varepsilon_0 = 0.1, \quad \alpha_0 = \beta_0 = 2^{-13},$$

with the following property. Choose, in this order, integers $k \geq 2$, $N \geq N_0$, and $n \geq 1$; a privacy pair (ε, δ) ; and an arbitrary randomized, computationally unrestricted, possibly improper map

$$A : (X_{k,N} \times \{0, 1\})^n \longrightarrow \mathcal{H}_{k,N}.$$

If Assumptions 1, 2, and 3 hold for this same candidate, then

$$n \geq ak \log_2^* N. \tag{1}$$

For the diagonal specialization, put $T_s = \text{Tot}_2(s)$ and

$$r_0 = \min\{s \in \mathbb{Z}_{\geq 2} : T_s \geq N_0\}.$$

For every integer $r \geq 2$,

$$\text{VC}(C_{r,T_r}) = r, \quad \text{LD}(C_{r,T_r}) = rT_{r-1}, \quad |C_{r,T_r}| = (T_r + 1)^r, \tag{2}$$

and

$$r - 1 \leq \log_2^* \text{LD}(C_{r,T_r}) \leq r. \tag{3}$$

Consequently,

$$\frac{1}{2}r^2 \leq \text{VC}(C_{r,T_r}) \log_2^* \text{LD}(C_{r,T_r}) \leq r^2. \tag{4}$$

For every $r \geq r_0$, every diagonal candidate with $k = r$, $N = T_r$ satisfying the same three assumptions obeys

$$n \geq ar^2 \quad \text{and} \quad n \geq a \text{VC}(C_{r,T_r}) \log_2^* \text{LD}(C_{r,T_r}). \tag{5}$$

All constants are fixed before $r, k, N, n, \varepsilon, \delta$, and A .

The deterministic comparison with the additive expression is

$$2r - 1 \leq \text{VC}(C_{r,T_r}) + \log_2^* \text{LD}(C_{r,T_r}) \leq 2r. \quad (6)$$

At the same time,

$$r^2 = o\left(\text{VC}(C_{r,T_r}) 2^{\log_2^* \text{LD}(C_{r,T_r})}\right) \quad \text{and} \quad r^2 = o(\log |C_{r,T_r}|). \quad (7)$$

Theorem 3.1 is a lower bound for this explicit class family in the stated candidate regime. It does not prove the stronger $\Omega(\text{VC} 2^{\log_2^* \text{LD}})$ milestone, an $\Omega(\log |C|)$ lower bound, a universal upper bound, or a matching combinatorial characterization for every class of finite Littlestone dimension. The comparisons in (7) concern expressions on the tower diagonal; they are not upper bounds on the true sample complexity.

4 Proof Sketch

The structural calculation is exact. Distinct cuts give $|C_{k,N}| = (N+1)^k$; one point from each tag is shattered, while two points from one tag cannot realize the labeling $(1, 0)$, so $\text{VC}(C_{k,N}) = k$. For Littlestone dimension, a one-tag binary-search tree has depth $\lfloor \log_2(N+1) \rfloor$. Concatenating one such tree per tag gives the lower bound, and summing the corresponding halving mistake budgets gives the matching upper bound; see Lemmas A.3–A.6 and Proposition A.1.

The one-block hard core comes from the unrestricted improper threshold lower bound of Alon et al. (2019). We first transport its ordered domain, labels, replacement adjacency, and population-risk convention exactly to $[N]$ and extract fixed constants in Proposition A.2. Markov’s inequality then converts the contrapositive high-probability statement into a strict expected-risk gap. The set of all finite source-private kernels is a compact polytope. Separating its continuous payoff image from the positive orthant yields the finite minimax prior of Proposition A.4; the proof includes the finite-dimensional separation and exact Caratheodory reduction rather than assuming a minimax swap.

We next calibrate the simulated budget $M = \max\{8, \lceil 4n/k \rceil\}$ under the strict negation of (1). For a hidden uniform coordinate J , the ideal product experiment has an independent k -fold prior, conditionally i.i.d. product data, and an exact selected-risk/product-risk identity even for tag-asymmetric improper learners. The hidden usage count is exactly $\text{Bin}(n, 1/k)$, and a two-case falling-factorial argument proves $\Pr\{U > M\} < 2^{-9}$, with zero overflow for $k = 2, 3$.

The simulator exposes an arbitrary size- M one-block input only through successive occurrences of tag J . Lemma A.31 proves pathwise that each input record enters at most one synthesized row. Thus a single input replacement changes at most one row passed to A . Keeping the learner’s coins random, then mixing over one common input-independent seed and restricting the output, preserves a single (ε, δ) privacy charge; this gives Proposition A.16. A shared-randomness coupling identifies actual and ideal datasets off overflow and loses only $\Pr\{U > M\}$ in expected selected risk.

Applying the finite hard prior to this certified simulator gives actual risk strictly above 2^{-8} . Coupling, the overflow bound, and the exact selected-risk identity yield ideal product risk strictly above 2^{-9} . Finite-support averaging extracts one deterministic product instance. The distribution-free PAC premise bounds the identical fixed-instance expectation by 2^{-12} , a contradiction. This discharges the strict negation and proves Proposition A.21. Finally, exact tower inversion and floor arithmetic give (2)–(7).

References

ALON, N., LIVNI, R., MALLIARIS, M. and MORAN, S. (2019). Private PAC learning implies finite littlestone dimension. Version 3. [4](#), [9](#)

A Proof Details

A.1 Exact class parameters

We use the following explicit convention. A complete binary instance tree of depth D over a domain $\mathcal{X}$ has an instance $x_\sigma \in \mathcal{X}$ at every node $\sigma \in \{0, 1\}^{<D}$. A binary class $H \subseteq \{0, 1\}^{\mathcal{X}}$ shatters the tree if, for every $b = (b_1, \dots, b_D) \in \{0, 1\}^D$, there is $h_b \in H$ such that

$$h_b(x_{b_{<r}}) = b_r \quad (1 \leq r \leq D),$$

where $b_{<r} = (b_1, \dots, b_{r-1})$. The Littlestone dimension is the largest shattered depth. This is the convention used throughout the proof; no tree/mistake equivalence is assumed without proof.

Lemma A.1 (Distinct tagged cuts and exact cardinality). *Under the primitive tagged-threshold definitions with integers $k \geq 2$ and $N \geq 2$, the functions τ_t , $t \in [N+1]$, are pairwise distinct, the parameterization $\mathbf{t} \mapsto c_{\mathbf{t}}$ is injective, and*

$$|C_{k,N}| = (N+1)^k.$$

Proof. If $t < s$ in $[N+1]$, then $t \leq N$, so $x = t$ is a valid point of $[N]$. At that point,

$$\tau_t(t) = 1, \quad \tau_s(t) = 0.$$

Thus all $N+1$ cuts are distinct. This comparison includes the endpoints: $\tau_1 \equiv 1$ and $\tau_{N+1} \equiv 0$ on $[N]$, and they disagree at every point.

If $\mathbf{t} \neq \mathbf{s}$, choose a tag i for which $t_i \neq s_i$. The preceding one-tag argument supplies an $x \in [N]$ with $\tau_{t_i}(x) \neq \tau_{s_i}(x)$, hence

$$c_{\mathbf{t}}(i, x) \neq c_{\mathbf{s}}(i, x).$$

The parameter map is therefore injective. It is surjective onto $C_{k,N}$ by the class definition, and its domain has size $(N+1)^k$, proving the formula. $\square$

Lemma A.2 (VC dimension of the tagged threshold product). *Under the primitive tagged-threshold definitions with integers $k \geq 2$ and $N \geq 2$,*

$$\text{VC}(C_{k,N}) = k.$$

Proof. For the lower bound, consider

$$S = \{(i, 1) : i \in [k]\}.$$

Given arbitrary desired labels $b_1, \dots, b_k \in \{0, 1\}$, set

$$t_i = \begin{cases} 1, & b_i = 1, \\ 2, & b_i = 0. \end{cases}$$

Both choices lie in $[N+1]$ because $N \geq 2$, and $\tau_{t_i}(1) = b_i$. Hence S is shattered and $\text{VC}(C_{k,N}) \geq k$.

For the upper bound, any set of $k+1$ distinct points of $[k] \times [N]$ contains two points (i, x) and (i, y) on the same tag. Relabel them so that $x < y$. For every threshold t_i ,

$$\tau_{t_i}(x) = 1 \implies x \geq t_i \implies y \geq t_i \implies \tau_{t_i}(y) = 1.$$

Thus the labeling $(1, 0)$ on these two points cannot be realized by any member of $C_{k,N}$. No $(k+1)$ -point set is shattered, so $\text{VC}(C_{k,N}) \leq k$. $\square$

Lemma A.3 (Exact one-tag Littlestone dimension and halving budget). *Under the primitive one-tag definition*

$$T_N = \{\tau_t : t \in [N+1]\} \subseteq \{0,1\}^{[N]},$$

with $N \geq 2$, and Lemma A.1, define

$$q = N+1, \quad d_N = \lfloor \log_2 q \rfloor.$$

Then T_N shatters a complete binary instance tree of depth d_N , shatters no tree of depth greater than d_N , and admits a deterministic online prediction rule that makes at most d_N mistakes on every finite labeled sequence realizable by T_N . Consequently,

$$\text{LD}(T_N) = d_N.$$

Proof. First, suppose T_N shatters a tree of depth D . Choose one witnessing concept for each of the 2^D root-to-leaf label sequences. Two different label sequences first diverge at some common node and prescribe opposite labels to the same queried point there. Their witnessing concepts must therefore be different. Lemma A.1 gives exactly q distinct one-tag concepts, so

$$2^D \leq q,$$

and hence $D \leq \lfloor \log_2 q \rfloor = d_N$.

For the matching lower bound, put $m = 2^{d_N}$, so $m \leq q = N+1$ and $m-1 \leq N$. We construct a depth- d_N tree using only the thresholds $t \in [m]$. At a node whose surviving threshold set is a consecutive interval

$$I = \{a, a+1, \dots, a+2^s-1\}$$

of size 2^s , where $s \geq 1$, query

$$x_I = a + 2^{s-1} - 1.$$

This is a valid point of $[N]$: throughout the construction $1 \leq x_I \leq m-1 \leq N$. On the edge labeled 1, retain the lower half $\{a, \dots, x_I\}$; on the edge labeled 0, retain the upper half $\{x_I+1, \dots, a+2^s-1\}$. In the first case every retained threshold obeys $t \leq x_I$ and therefore $\tau_t(x_I) = 1$; in the second case every retained threshold obeys $t > x_I$ and therefore $\tau_t(x_I) = 0$. Starting from $I = [m]$, every root-to-leaf path leaves a singleton after exactly d_N splits, and that singleton threshold realizes every edge label on the path. The constructed tree is therefore shattered.

It remains to record the online interface needed for the product upper bound. Maintain the version space $V \subseteq T_N$ of concepts consistent with all previously revealed examples, starting from $V = T_N$. At a query x , predict a label having the larger number of concepts in

$$V_y(x) = \{\tau \in V : \tau(x) = y\}, \quad y \in \{0,1\},$$

with an arbitrary fixed tie rule. After the true label y is revealed, replace V by $V_y(x)$. On a realizable sequence, the realizing threshold remains in V , so V never becomes empty. Whenever the prediction is wrong, the surviving side has cardinality at most half the previous cardinality. If r mistakes occur, then

$$1 \leq |V| \leq q 2^{-r},$$

so $2^r \leq q$ and therefore $r \leq d_N$. This also covers the endpoint targets $t = 1$ and $t = N+1$: each belongs to the initial version space and remains present whenever it is the realizing target. $\square$

Lemma A.4 (Concatenated tagged trees give the product lower bound). *Under the primitive definition of $C_{k,N}$, with $k \geq 2, N \geq 2$, and the depth- d_N one-tag tree from Lemma A.3, the product class shatters a complete tree of depth kd_N . Hence*

$$\text{LD}(C_{k,N}) \geq kd_N.$$

Proof. Attach a copy of the one-tag tree on tag 1 first. Below every leaf attach a copy on tag 2, and continue through tag k . Equivalently, split any path label vector $b \in \{0, 1\}^{kd_N}$ into consecutive blocks

$$b = (b^{(1)}, \dots, b^{(k)}), \quad b^{(i)} \in \{0, 1\}^{d_N}.$$

During block i , if the corresponding one-tag copy queries $x \in [N]$, the product tree queries (i, x) . By Lemma A.3, for each block $b^{(i)}$ there is a threshold $t_i \in [N + 1]$ whose one-tag labels realize that entire block. Because the coordinates $t_1, \dots, t_k$ are independent in the defining parameter set $[N + 1]^k$, the single product concept $c_{(t_1, \dots, t_k)}$ realizes all k blocks of the chosen path. Every length- kd_N path is therefore realized, proving the stated lower bound. $\square$

Lemma A.5 (Summed per-tag realizable mistake budget). *Under the primitive definition of $C_{k,N}$, with $k \geq 2, N \geq 2$, and the one-tag online rule of Lemma A.3, there is a deterministic online rule that makes at most kd_N mistakes on every finite sequence realizable by $C_{k,N}$.*

Proof. Maintain k independent one-tag version spaces $V_1, \dots, V_k$, each initialized to T_N . When the current instance is (i, x) , use the one-tag halving rule based only on V_i , reveal the resulting prediction, and update only V_i after observing the label.

If the full sequence is realized by c_t , then the subsequence on tag i is realized by τ_{t_i} . Lemma A.3 therefore bounds the number M_i of mistakes charged to tag i by d_N . Each global mistake occurs on exactly one tag and is charged exactly once, so

$$M_{\text{total}} = \sum_{i=1}^k M_i \leq \sum_{i=1}^k d_N = kd_N.$$

There is no cross-tag condition to maintain: observations on tag i constrain only t_i . The same argument includes $k = 2$ and includes any coordinate whose realizing threshold is either endpoint. $\square$

Lemma A.6 (A realizable mistake budget upper-bounds Littlestone dimension). *Let $H \subseteq \{0, 1\}^{\mathcal{X}}$ be any binary class and let $B \in \mathbb{Z}_{\geq 0}$. If a deterministic online prediction rule makes at most B mistakes on every finite labeled sequence realizable by H , then*

$$\text{LD}(H) \leq B.$$

Proof. Suppose instead that H shatters a complete tree of depth $B + 1$. Starting at its root, present the instance at the current node to the rule. If the rule predicts $p \in \{0, 1\}$, reveal the label $1 - p$ and descend along that edge. Repeat for all $B + 1$ levels. By construction the rule makes a mistake at every level. By shattering, the resulting root-to-leaf labeled sequence is realized by some single member of H . This is a realizable sequence with $B + 1$ mistakes, contradicting the assumed budget. $\square$

Proposition A.1 (Exact Littlestone dimension of the tagged product). *Under the primitive tagged-threshold definitions with integers $k \geq 2, N \geq 2$, and Lemmas A.4, A.5, and A.6,*

$$\text{LD}(C_{k,N}) = k \lfloor \log_2(N + 1) \rfloor.$$

Proof. Lemma A.4 gives

$$\text{LD}(C_{k,N}) \geq kd_N.$$

Lemma A.5 constructs a deterministic rule with realizable mistake budget kd_N . Applying Lemma A.6 with $H = C_{k,N}$ and $B = kd_N$ gives

$$\text{LD}(C_{k,N}) \leq kd_N.$$

The bounds agree, and $d_N = \lfloor \log_2(N+1) \rfloor$. $\square$

Together, Lemmas A.1 and A.2 and Proposition A.1 establish all three exact structural identities used below, including both endpoint cuts.

A.2 The unrestricted one-block threshold lower bound

For a randomized one-block learner $B : ([N] \times \{0, 1\})^M \rightarrow \{0, 1\}^{[N]}$, write (*W-PAC*) for

$$\Pr_{\substack{S \sim (Q^{\tau_t})^M \\ g \sim B(S)}} \left\{ R_Q(g, \tau_t) \leq \frac{1}{16} \right\} \geq \frac{15}{16} \quad \text{for every } (t, Q) \in \mathcal{I}_N,$$

and write (*W-LB*) for $M \geq b_* \log_2^* N$. The external input is the unrestricted improper threshold theorem of Alon et al. (2019); all convention and parameter transport is proved here.

Lemma A.7 (Fixed-constant form of the ALMM v3 threshold theorem). *The threshold theorem of Alon et al. (2019, Theorem 1), together with its PAC and privacy definitions and the fixed constants in its proof, implies that there exist universal constants*

$$b_0 > 0, \quad d_0 > 0, \quad N_0 \in \mathbb{Z}_{\geq 2}$$

such that, for every finite ordered real domain X with $|X| = N \geq N_0$, every integer $M \geq 8$, and every possibly randomized, possibly improper threshold learner on X that is $(1/16, 1/16)$ -accurate in the source PAC sense and

$$\left(0.1, \frac{d_0}{M^2 \log M} \right)\text{-differentially private,}$$

one has

$$M \geq b_0 \log_2^* N.$$

Proof. The source theorem states

$$M = \Omega(\log^* N) \quad \text{under} \quad \varepsilon = 0.1, \quad \delta = O\left(\frac{1}{M^2 \log M}\right). \quad (\text{S2.1})$$

The constants in both asymptotic symbols are absolute: the theorem fixes the accuracy and privacy constants before quantifying over the domain and learner. This is also visible in the active proof. Its private homogeneous-set lemma uses the explicit empirical-size cap

$$\delta \leq \frac{1}{10^3 m^2 \log m}, \quad (\text{S2.2})$$

and the source's constant-factor empirical reduction increases the original PAC sample size by at most the absolute factor nine while preserving accuracy and privacy.

For completeness, the factor-nine change does not leave an unverified denominator. The function

$$x \mapsto \frac{\log x}{\log(9x)}$$

is increasing for $x > 1$, because its derivative is

$$\frac{\log 9}{x(\log(9x))^2} > 0.$$

Hence, for every $M \geq 8$,

$$\frac{\log M}{\log(9M)} \geq \frac{\log 8}{\log 72}. \quad (\text{S2.3})$$

Choose d_0 no larger than the fixed privacy constant represented by the $O(\cdot)$ in (S2.1), and shrink it further, if needed, so that

$$0 < d_0 \leq \frac{\log 8}{81000 \log 72}. \quad (\text{S2.4})$$

Then it obeys

$$\frac{d_0}{M^2 \log M} \leq \frac{1}{10^3 (9M)^2 \log(9M)}. \quad (\text{S2.5})$$

If the empirical reduction uses a size $m \leq 9M$, then $m^2 \log m \leq (9M)^2 \log(9M)$, so (S2.5) also implies (S2.2) at that actual size. Thus the source proof exhibits a strictly positive universal privacy constant after its own constant-factor reduction; (S2.4) is only a conservative witness and is not claimed as a numerical constant printed in the theorem statement.

Unpacking the Ω -conclusion in (S2.1) gives an absolute lower-bound constant and a finite domain threshold. The source's unadorned iterated-log convention and the branch's base-two convention differ by at most an absolute additive number of iterations: for some universal K_{base} ,

$$\log_{\text{source}}^* N \geq \log_2^* N - K_{\text{base}}. \quad (\text{S2.6})$$

This is the standard fixed-base comparison: multiplying a logarithm by the fixed base-change constant can require only a fixed number of additional iterates before entering $(0, 1]$. Increase the finite domain threshold so that

$$\log_2^* N \geq 2K_{\text{base}}$$

throughout the retained regime. Then (S2.6) is at least $\frac{1}{2} \log_2^* N$, and halving the source lower-bound constant gives a universal $b_0 > 0$. Absorb every finite exceptional domain and every fixed source convention into one integer $N_0 \geq 2$. This proves the displayed fixed-constant form for all $M \geq 8$. No value of b_0 or N_0 beyond existence is asserted. $\square$

Lemma A.8 (Exact order, label, PAC, output, and adjacency transport). *Under the primitive one-block definitions with $N \geq 2$, and consistently with the endpoint/distinctness certificate in Lemma A.1, let*

$$B : ([N] \times \{0, 1\})^M \longrightarrow \{0, 1\}^{[N]}$$

be any randomized map. There is a rowwise bijective transport to a possibly randomized, unrestricted ALMM learner

$$\tilde{B} : (X \times \{-1, +1\})^M \longrightarrow \{-1, +1\}^X$$

on an N -point ordered real set X such that:

1. All increasing thresholds, including the all-positive and all-negative endpoint restrictions, correspond to $\{\tau_t : t \in [N + 1]\}$.
2. Every arbitrary output of B corresponds bijectively to an arbitrary output of $\tilde{B}$, with no properness or monotonicity.
3. For every t, Q , the labeled sample laws are exact i.i.d. products of the same fixed size M , and population risks agree pointwise.
4. Two input datasets differ in one labeled row by replacement before transport if and only if their images do so after transport.
5. (W-PAC) for every t, Q is exactly the source realizable-PAC premise for $\tilde{B}$.

The same conclusions hold if a presentation of the source reverses the threshold orientation, after reversing the order or complementing all labels.

Proof. Choose any increasing real sequence

$$x_1 < x_2 < \cdots < x_N$$

and put $X = \{x_1, \dots, x_N\}$. Let $\phi : [N] \rightarrow X$ be the order bijection $\phi(j) = x_j$, and let

$$\lambda(-1) = 0, \quad \lambda(+1) = 1.$$

For the increasing source threshold with cut index t , write

$$s_t(x_j) = \begin{cases} -1, & j < t, \\ +1, & j \geq t. \end{cases}$$

Then, for every $j \in [N]$ and $t \in [N + 1]$,

$$\lambda(s_t(\phi(j))) = \mathbf{1}\{j \geq t\} = \tau_t(j). \tag{S2.7}$$

The endpoint $t = 1$ is all positive/all one, and $t = N + 1$ is all negative/all zero. They are restrictions of real half-line thresholds with the cut at or below x_1 , respectively above x_N . Lemma A.1 independently certifies that these are legal, distinct branch members. Even under a convention that lists only interior cut representatives, that source family is a subclass of the branch family, so a learner satisfying (W-PAC) still meets every source target requirement.

Transport a source labeled row by

$$(x_j, y) \longmapsto (j, \lambda(y)). \tag{S2.8}$$

For a source sample S , apply (S2.8) row by row to obtain $S^\flat$, run $B(S^\flat)$, and define

$$\tilde{B}(S)(x_j) = \lambda^{-1}\left(B(S^\flat)(j)\right). \tag{S2.9}$$

Equation (S2.9) is defined for every branch output, including constant, oscillating, and non-monotone functions. It transports the internal randomness unchanged and uses no projection onto the target class.

Let Q be a probability law on $[N]$, and let $\tilde{Q} = \phi_{\#}Q$ be its pushforward to X . For every deterministic branch output g , with transported source output $\tilde{g}(x_j) = \lambda^{-1}(g(j))$, (S2.7) gives

$$\begin{aligned} L_{\tilde{Q}^{s_t}}(\tilde{g}) &= \Pr_{J \sim Q}[\tilde{g}(x_J) \neq s_t(x_J)] \\ &= \Pr_{J \sim Q}[g(J) \neq \tau_t(J)] \\ &= R_Q(g, \tau_t). \end{aligned} \tag{S2.10}$$

The row transport is deterministic and coordinatewise, so

$$S \sim (\tilde{Q}^{s_t})^M \iff S^b \sim (Q^{\tau_t})^M. \tag{S2.11}$$

There is no random or expected sample-size substitution. Every distribution on labeled source examples realizable by a threshold equals $\tilde{Q}^{s_t}$ for its feature marginal $\tilde{Q}$ and its realizing cut (up to changes on zero-mass points). Thus (S2.10)–(S2.11), including the learner’s unchanged randomness, identify (W-PAC) with the source PAC premise for every realizable threshold distribution.

Finally, (S2.8) is a bijection on individual labeled rows and is applied without mixing row indices. Consequently two length- M samples are equal or differ at exactly one row before transport exactly when the corresponding samples are equal or differ at that same row afterward. Output relabeling in (S2.9) is bijective postprocessing. Hence the replacement-DP event inequalities are identical under the event bijection. The source’s two-sided indistinguishability is equivalent to the branch’s one-sided display quantified over all ordered adjacent pairs, because adjacency is symmetric.

If a threshold convention uses positive labels below the cut, either replace λ by the complemented label bijection or replace $\phi(j)$ by the order-reversing bijection x_{N+1-j} . These are still rowwise bijections, so (S2.10)–(S2.11) and adjacency preservation remain exact. $\square$

Lemma A.9 (Privacy monotonicity at the source cap). *Under the fixed-size replacement convention, let $M \geq 8$, $d_0 > 0$, $0 \leq \varepsilon \leq 0.1$, and*

$$0 \leq \delta \leq \frac{d_0}{M^2 \log M}.$$

If a randomized kernel is (ε, δ) -differentially private, then it is also

$$\left(0.1, \frac{d_0}{M^2 \log M}\right)\text{-differentially private.}$$

This conclusion includes $M = 8$, $\varepsilon < 0.1$, δ strictly below the cap, and $\delta = 0$.

Proof. For adjacent datasets S, S' and every output event E , privacy gives

$$\Pr[B(S) \in E] \leq e^\varepsilon \Pr[B(S') \in E] + \delta.$$

Since the event probability is nonnegative, $e^\varepsilon \leq e^{0.1}$, and $\delta \leq d_0/(M^2 \log M)$,

$$\Pr[B(S) \in E] \leq e^{0.1} \Pr[B(S') \in E] + \frac{d_0}{M^2 \log M}. \tag{S2.12}$$

Swapping S, S' gives the reverse source inequality. There is one application of privacy at the same sample size; no group privacy, composition, subsampling, or adjacency conversion occurs. At $M = 8$, $M^2 \log M = 64 \log 8 > 0$, so the cap in (S2.12) is finite and positive. Decreasing either privacy parameter only strengthens (S2.12). $\square$

Proposition A.2 (Current-notation unrestricted ALMM threshold wrapper). *Under Lemmas A.7, A.8, and A.9, there are universal $b_*, d_* > 0$ and $N_* \geq 2$ such that, for every $N \geq N_*$, integer $M \geq 8$, $0 \leq \varepsilon \leq 0.1$, and $0 \leq \delta \leq d_*/(M^2 \log M)$, every randomized unrestricted map*

$$B : ([N] \times \{0, 1\})^M \rightarrow \{0, 1\}^{[N]}$$

that is (ε, δ) -DP under one-row replacement and satisfies (W-PAC) for every $t \in [N + 1]$ and every law Q on $[N]$ must satisfy (W-LB):

$$M \geq b_* \log_2^* N.$$

Proof. Take

$$b_* = b_0, \quad d_* = d_0, \quad N_* = N_0$$

from Lemma A.7. Lemma A.9 promotes the stated privacy guarantee to

$$\left(0.1, \frac{d_0}{M^2 \log M}\right)\text{-DP}.$$

Apply the exact row/order/label transport of Lemma A.8. It produces a possibly randomized source learner whose output ranges over all $\{\pm 1\}$ -valued functions, whose input consists of exactly M iid labeled examples, whose loss is the identical population 0-1 risk, and whose adjacency/privacy guarantee is the identical one-entry replacement guarantee. By (W-PAC) and (S2.10)–(S2.11), this transported learner is $(1/16, 1/16)$ -accurate for every source threshold-realizable distribution. All hypotheses of Lemma A.7 are therefore discharged, so

$$M \geq b_0 \log_2^* N = b_* \log_2^* N.$$

The proof does not select a hard instance and does not assert expected-risk hardness. Those are conclusions of later contrapositive and minimax steps, not outputs of this proposition. $\square$

Lemmas A.7–A.9 yield Proposition A.2, the precise one-block implication used in the remainder of the proof.

A.3 A strict expected-loss hard instance

Lemma A.10 (Expectation-to-PAC conversion at the ALMM constants). *Under the primitive one-block definitions, fix integers $N \geq 2$ and $M \geq 1$, and let*

$$B : ([N] \times \{0, 1\})^M \rightarrow \{0, 1\}^{[N]}$$

be any randomized unrestricted map. If, for every $t \in [N + 1]$ and every probability law Q on $[N]$,

$$\mathbb{E}_{\substack{S \sim (Q^{\tau_t})^M \\ g \sim B(S)}} R_Q(g, \tau_t) \leq 2^{-8}, \tag{S3.1}$$

then, for every such t, Q ,

$$\Pr_{\substack{S \sim (Q^{\tau_t})^M \\ g \sim B(S)}} \left[R_Q(g, \tau_t) > \frac{1}{16} \right] \leq \frac{1}{16}. \tag{S3.2}$$

Thus B satisfies (W-PAC).

Proof. Fix an arbitrary $t \in [N+1]$ and arbitrary law Q on $[N]$, while keeping the same fixed sample size M . Under the joint randomness of $S \sim (Q^{\tau_t})^M$ and $g \sim B(S)$, define the proof-local random variable

$$Z_{t,Q} := R_Q(g, \tau_t).$$

For every realized output g , population 0-1 risk is a probability, so

$$0 \leq Z_{t,Q} \leq 1. \quad (\text{S3.3})$$

Apply Markov's inequality at the positive threshold $1/16$. The strict failure event in (W-PAC) is contained in the weak-threshold event, hence

$$\begin{aligned} \Pr\left(Z_{t,Q} > \frac{1}{16}\right) &\leq \Pr\left(Z_{t,Q} \geq \frac{1}{16}\right) \\ &\leq 16 \mathbb{E} Z_{t,Q} \\ &\leq 16 \cdot 2^{-8} \\ &= 2^{-4} = \frac{1}{16}. \end{aligned} \quad (\text{S3.4})$$

The final inequality remains valid when the expectation in (S3.1) equals 2^{-8} ; no strict inequality has been inserted. Since (t, Q) was arbitrary, (S3.4) holds simultaneously as a universally quantified statement for every fixed (t, Q) , which is exactly (W-PAC). Randomization and improperness of g do not affect (S3.3). $\square$

Proposition A.3 (Expected-loss hardness below the one-block threshold bound). *Let $b_*, d_* > 0$ and $N_* \geq 2$ be the constants supplied by Proposition A.2. Under that proposition and Lemma A.10, for every pair of integers $N \geq N_*$, $M \geq 8$ satisfying*

$$M < b_* \log_2^* N,$$

every randomized unrestricted map

$$B : ([N] \times \{0, 1\})^M \rightarrow \{0, 1\}^{[N]}$$

that is $(0.1, d_/(M^2 \log M))$ -DP under one-row replacement admits some $t \in [N+1]$ and some probability law Q on $[N]$ for which*

$$\mathbb{E}_{\substack{S \sim (Q^{\tau_t})^M \\ g \sim B(S)}} R_Q(g, \tau_t) > 2^{-8}. \quad (\text{S3.5})$$

Proof. Fix N, M and an arbitrary randomized unrestricted B satisfying all the displayed hypotheses. Suppose (S3.5) were false. The exact logical negation of the existential strict inequality is

$$\text{for every } t \in [N+1] \text{ and every law } Q \text{ on } [N], \quad \mathbb{E}_{\substack{S \sim (Q^{\tau_t})^M \\ g \sim B(S)}} R_Q(g, \tau_t) \leq 2^{-8}. \quad (\text{S3.6})$$

Lemma A.10 applies to the same map B , the same exact sample size M , and every fixed (t, Q) . It converts (S3.6) into (W-PAC), including the allowed equality $\Pr(R_Q(g, \tau_t) > 1/16) = 1/16$.

All remaining hypotheses of Proposition A.2 now hold: $N \geq N_*$, integer $M \geq 8$, the map is arbitrary and improper, and its privacy parameters equal the permitted endpoints

$$\varepsilon = 0.1, \quad \delta = \frac{d_*}{M^2 \log M}.$$

The denominator is positive even at $M = 8$, since $\log 8 > 0$. The wrapper therefore yields

$$M \geq b_* \log_2^* N,$$

contradicting the strict hypothesis $M < b_* \log_2^* N$. Hence (S3.6) is false, so a fixed pair (t, Q) satisfying (S3.5) exists. The pair may depend on the fixed learner B , because only worst-instance hardness is asserted here; no learner-independent prior or minimax conclusion is asserted. $\square$

The Markov conversion in Lemma A.10 and the contrapositive of Proposition A.2 give Proposition A.3 with strict gap $\eta = 2^{-8}$.

A.4 A finite public minimax prior

Let $b_*, d_* > 0$ and $N_* \geq 2$ be the constants from Proposition A.3, and set $\eta = 2^{-8}$. Throughout this subsection the scalar hard regime is explicitly

$$N \geq N_*, \quad M \in \mathbb{Z}_{\geq 8}, \quad M < b_* \log_2^* N. \quad (\text{S4.H})$$

For these fixed N, M , the interface to be constructed is one finite law $\mu_{N,M}$, selected before the learner, such that every randomized unrestricted one-block map $B : ([N] \times \{0, 1\})^M \rightarrow \{0, 1\}^{[N]}$ satisfying the source privacy cap obeys the implication

$$\left[B \text{ is } \left(0.1, \frac{d_*}{M^2 \log M} \right)\text{-DP} \right] \implies \mathbb{E}_{(t,Q) \sim \mu_{N,M}} \mathbb{E}_{\substack{S \sim (Q^{\tau_t})^M \\ g \sim B(S)}} R_Q(g, \tau_t) > \eta. \quad (\text{S4.HP})$$

The quantifier order in (S4.HP) is $\exists \mu_{N,M} \forall B$. The objects $\mathcal{K}_{N,M}$, $\mathcal{I}_N$, and their payoff are all finite dimensional, although $\mathcal{I}_N$ itself contains a continuum of distributions.

Lemma A.11 (Exact compact polytope of source-private kernels). *Under the primitive finite one-block definitions, let $d_* > 0$, fix integers $N \geq 2$, $M \geq 8$, and put $\bar{\delta}_M = d_*/(M^2 \log M)$. Define*

$$\mathcal{S}_{N,M} := ([N] \times \{0, 1\})^M, \quad \mathcal{G}_N := \{0, 1\}^{[N]}, \quad D := |\mathcal{S}_{N,M}| |\mathcal{G}_N| = (2N)^M 2^N.$$

Let $\mathcal{K}_{N,M} \subset \mathbb{R}^D$ consist of the arrays $K = (K_{s,g})_{s \in \mathcal{S}_{N,M}, g \in \mathcal{G}_N}$ satisfying

$$K_{s,g} \geq 0, \quad \sum_{g \in \mathcal{G}_N} K_{s,g} = 1 \quad (s \in \mathcal{S}_{N,M}), \quad (\text{S4.6})$$

and, for every ordered adjacent pair $s \simeq s'$ and every event $E \subseteq \mathcal{G}_N$,

$$\sum_{g \in E} K_{s,g} \leq e^{0.1} \sum_{g \in E} K_{s',g} + \bar{\delta}_M. \quad (\text{S4.7})$$

Then $\mathcal{K}_{N,M}$ is a nonempty compact convex polytope, and its points are in exact bijection with the output kernels of randomized unrestricted $(0.1, \bar{\delta}_M)$ -DP maps $B : \mathcal{S}_{N,M} \rightarrow \mathcal{G}_N$.

Proof. Both $\mathcal{S}_{N,M}$ and $\mathcal{G}_N$ are finite. Consequently, (S4.6) is a finite family of linear equalities and inequalities. Although the all-event privacy family is large, it is also finite: there are finitely many ordered adjacent pairs and exactly $2^{|\mathcal{G}_N|}$ subsets of the finite output space. Thus (S4.7) is the *exact* all-event approximate-DP definition expressed as finitely many closed affine halfspaces. In particular, no invalid replacement of all-event privacy by singleton-only constraints is made.

The product of row simplexes defined by (S4.6) is closed and bounded in $\mathbb{R}^D$, hence compact. Intersecting it with the finitely many closed halfspaces (S4.7) leaves a compact set. All constraints are affine, so the intersection is convex and is a polytope.

For nonemptiness, fix any $g_0 \in \mathcal{G}_N$ and set

$$K_{s,g}^0 := \mathbf{1}\{g = g_0\} \quad \text{for every } s, g.$$

The output distribution is identical for s and s' . Hence it is $(0,0)$ -DP and satisfies (S4.7), including the events containing g_0 and those not containing it. Thus $K^0 \in \mathcal{K}_{N,M}$.

If B is any randomized map, define $K_{s,g} = \Pr[B(s) = g]$. Row-stochasticity is (S4.6), and its all-event DP inequality is exactly (S4.7), so a source-private B gives a point of $\mathcal{K}_{N,M}$. Conversely, given $K \in \mathcal{K}_{N,M}$, on input s draw g from the finite row distribution $(K_{s,g})_g$. This defines a randomized unrestricted map B_K , and (S4.7) proves its exact source-cap privacy for every event. The full set $\mathcal{G}_N$ includes every improper, nonmonotone hypothesis.

Only the output law on each input matters to both privacy and the one-call expected loss. Therefore arbitrary internal randomization is encoded without loss. Also, if a public object is hardwired into a learner's code, the resulting map still has one such finite kernel and belongs to $\mathcal{K}_{N,M}$ whenever it satisfies source privacy. If a map is (ε, δ) -DP with $0 \leq \varepsilon \leq 0.1$ and $0 \leq \delta \leq \bar{\delta}_M$, its event inequality is at most the right side of (S4.7), so its kernel also belongs to $\mathcal{K}_{N,M}$. Thus the polytope includes the exact cap and every stronger privacy pair below it. $\square$

Lemma A.12 (Continuous payoff geometry and uniform strict game value). *Under Lemma A.11 and Proposition A.3, fix integers N, M satisfying (S4.H). Give $[N+1]$ the discrete topology and $\Delta([N])$ its Euclidean simplex topology, and set $\mathcal{I}_N = [N+1] \times \Delta([N])$. Then:*

1. $\mathcal{I}_N$ is compact.
2. The expected one-block loss $\ell_{N,M}(K; t, Q)$ admits the finite coefficient representation

$$\ell_{N,M}(K; t, Q) = \langle K, a_{N,M}(t, Q) \rangle.$$

It is affine and continuous in K , and jointly continuous in (K, t, Q) . For fixed K, t , its dependence on Q is generally polynomial of degree at most $M+1$, so only continuity, not affinity in (t, Q) , is claimed.

3. The function

$$f_{N,M}(K) := \max_{(t,Q) \in \mathcal{I}_N} \ell_{N,M}(K; t, Q)$$

is continuous on $\mathcal{K}_{N,M}$, and the attained value

$$v_{N,M} := \min_{K \in \mathcal{K}_{N,M}} f_{N,M}(K) \tag{S4.8}$$

satisfies $v_{N,M} > \eta$.

Proof. The simplex $\Delta([N])$ is closed and bounded in $\mathbb{R}^N$, and $[N+1]$ is finite. Their product $\mathcal{I}_N$ is compact. This includes every boundary face and every point mass of the simplex.

Write an arbitrary labeled dataset as

$$s = ((x_1, y_1), \dots, (x_M, y_M)) \in \mathcal{S}_{N,M}.$$

For $(t, Q) \in \mathcal{I}_N$ and $g \in \mathcal{G}_N$, define

$$p_{t,Q}(s) := \left(\prod_{j=1}^M Q(x_j) \right) \mathbf{1}\{y_j = \tau_t(x_j) \text{ for every } j\}, \quad (\text{S4.9})$$

and

$$r_{t,Q}(g) := R_Q(g, \tau_t) = \sum_{x=1}^N Q(x) \mathbf{1}\{g(x) \neq \tau_t(x)\}. \quad (\text{S4.10})$$

Thus $p_{t,Q}$ is exactly the mass function of $(Q^{\tau_t})^M$, including zero mass on every label-inconsistent dataset. Define the coefficient vector $a_{N,M}(t, Q) \in \mathbb{R}^D$ by

$$a_{N,M}(t, Q)_{s,g} := p_{t,Q}(s) r_{t,Q}(g). \quad (\text{S4.11})$$

For the learner represented by K , direct expansion of the sample and output expectations gives

$$\begin{aligned} \ell_{N,M}(K; t, Q) &:= \mathbb{E}_{\substack{S \sim (Q^{\tau_t})^M \\ g \sim B_K(S)}} R_Q(g, \tau_t) \\ &= \sum_{s \in \mathcal{S}_{N,M}} p_{t,Q}(s) \sum_{g \in \mathcal{G}_N} K_{s,g} r_{t,Q}(g) \\ &= \sum_{s,g} K_{s,g} a_{N,M}(t, Q)_{s,g} = \langle K, a_{N,M}(t, Q) \rangle. \end{aligned} \quad (\text{S4.12})$$

For fixed t , every coordinate in (S4.11) is a polynomial in the coordinates of Q : (S4.9) has degree M and (S4.10) has degree one. Hence $a_{N,M}$ is continuous on each simplex copy and therefore on the finite disjoint union $\mathcal{I}_N$. Equation (S4.12) is linear, hence affine and continuous, in K , and is jointly continuous in (K, t, Q) . The product structure in (S4.9) is precisely why no affine dependence on raw Q is asserted or needed.

The compact coefficient image $a_{N,M}(\mathcal{I}_N)$ is bounded. Therefore, for any $K, K' \in \mathcal{K}_{N,M}$,

$$\begin{aligned} |f_{N,M}(K) - f_{N,M}(K')| &\leq \max_{(t,Q) \in \mathcal{I}_N} |\langle K - K', a_{N,M}(t, Q) \rangle| \\ &\leq \|K - K'\|_2 \max_{(t,Q) \in \mathcal{I}_N} \|a_{N,M}(t, Q)\|_2. \end{aligned} \quad (\text{S4.13})$$

Thus $f_{N,M}$ is continuous. Its inner maximum is attained because $\mathcal{I}_N$ is compact, and its minimum in (S4.8) is attained because $\mathcal{K}_{N,M}$ is compact.

Now fix any $K \in \mathcal{K}_{N,M}$. By Lemma A.11, K defines an arbitrary randomized unrestricted source-private map B_K . Proposition A.3 therefore supplies some $(t, Q) \in \mathcal{I}_N$ such that

$$\ell_{N,M}(K; t, Q) > \eta.$$

Hence

$$f_{N,M}(K) > \eta \quad \text{for every } K \in \mathcal{K}_{N,M}. \quad (\text{S4.14})$$

Let K_* attain the minimum in (S4.8). Applying (S4.14) to this actual minimizer, rather than taking an unattained infimum, yields

$$v_{N,M} = f_{N,M}(K_*) > \eta. \quad (\text{S4.15})$$

This is the required compactness upgrade from learner-by-learner strict hardness to one uniform strict game value. $\square$

Lemma A.13 (Finite-dimensional strong separation). *Let $A \subset \mathbb{R}^m$ be nonempty, compact, and convex, and let $D \subset \mathbb{R}^m$ be nonempty, closed, and convex. If $A \cap D = \emptyset$, then there are a nonzero $\lambda \in \mathbb{R}^m$ and $c \in \mathbb{R}$ such that, after orienting λ toward A ,*

$$\sup_{z \in D} \langle \lambda, z \rangle < c < \inf_{u \in A} \langle \lambda, u \rangle.$$

Proof. The Euclidean distance between A and D is positive. Indeed, otherwise there would be $a_j \in A$ and $z_j \in D$ with $\|a_j - z_j\|_2 \rightarrow 0$; compactness gives a subsequence $a_j \rightarrow a_0 \in A$, hence $z_j \rightarrow a_0$, and closedness would put $a_0 \in D$. The distance is attained at some $(a_0, z_0) \in A \times D$: a minimizing sequence in A has a convergent subsequence, and the corresponding points of D are bounded and therefore have a convergent subsequence whose limit lies in D .

Put $\lambda = a_0 - z_0 \neq 0$. For every $a \in A$, convexity permits the segment $a_0 + t(a - a_0)$, and minimality at $t = 0$ gives

$$\langle \lambda, a - a_0 \rangle \geq 0.$$

Likewise, for every $z \in D$, the segment $z_0 + t(z - z_0)$ and minimality give

$$\langle \lambda, z - z_0 \rangle \leq 0.$$

Consequently

$$\sup_{z \in D} \langle \lambda, z \rangle \leq \langle \lambda, z_0 \rangle < \langle \lambda, a_0 \rangle \leq \inf_{a \in A} \langle \lambda, a \rangle,$$

where the strict gap is $\langle \lambda, a_0 - z_0 \rangle = \|\lambda\|_2^2 > 0$. Any scalar strictly between the middle two quantities is a valid c . $\square$

Lemma A.14 (Compact bilinear minimax via finite-dimensional separation). *Let X and Y be nonempty compact convex subsets of finite-dimensional real vector spaces, and let $p : X \times Y \rightarrow \mathbb{R}$ be continuous and affine in each argument (in particular, a restricted bilinear pairing). Then both outer extrema below are attained and*

$$\min_{x \in X} \max_{y \in Y} p(x, y) = \max_{y \in Y} \min_{x \in X} p(x, y). \quad (\text{S4.16})$$

Proof. Continuity on the compact product implies that $x \mapsto \max_{y \in Y} p(x, y)$ is continuous; for example, uniform continuity of p bounds the change of the maximum by the same uniform modulus. Hence

$$\alpha := \min_{x \in X} \max_{y \in Y} p(x, y)$$

is attained. The elementary weak minimax inequality gives

$$\max_{y \in Y} \min_{x \in X} p(x, y) \leq \alpha. \quad (\text{S4.17})$$

We prove the reverse inequality by separation. Fix any real $r < \alpha$. For each $x \in X$, compactness of Y supplies some $y_x \in Y$ with $p(x, y_x) > r$. Continuity in x gives a neighborhood on which this same strict inequality holds. Compactness of X yields a finite subcover, so there are $y_1, \dots, y_m \in Y$ such that

$$\max_{1 \leq j \leq m} p(x, y_j) > r \quad \text{for every } x \in X. \quad (\text{S4.18})$$

Define the affine continuous map

$$F : X \rightarrow \mathbb{R}^m, \quad F(x) = (p(x, y_1), \dots, p(x, y_m)),$$

and the closed convex lower orthant

$$\mathcal{D}_r := (-\infty, r]^m.$$

The image $F(X)$ is compact and convex. Equation (S4.18) says that

$$F(X) \cap \mathcal{D}_r = \emptyset.$$

Applying Lemma A.13 provides a nonzero $\lambda \in \mathbb{R}^m$ oriented so that

$$\sup_{z \in \mathcal{D}_r} \langle \lambda, z \rangle < \inf_{x \in X} \langle \lambda, F(x) \rangle. \quad (\text{S4.19})$$

Every coordinate of λ must be nonnegative: if $\lambda_j < 0$, sending the j -th coordinate of $z \in \mathcal{D}_r$ to $-\infty$ makes the left side of (S4.19) infinite. Normalize the nonzero nonnegative vector so that $\sum_j \lambda_j = 1$. Then the left side of (S4.19) is exactly r , attained at $(r, \dots, r)$, and hence

$$\inf_{x \in X} \sum_{j=1}^m \lambda_j p(x, y_j) > r. \quad (\text{S4.20})$$

Convexity of Y puts $y_r := \sum_j \lambda_j y_j$ in Y , while affinity in the second argument turns (S4.20) into

$$\min_{x \in X} p(x, y_r) > r. \quad (\text{S4.21})$$

Choose $r_q \uparrow \alpha$. Compactness of Y gives a convergent subsequence $y_{r_q} \rightarrow y_* \in Y$. For every fixed $x \in X$, (S4.21) and continuity give

$$p(x, y_*) = \lim_q p(x, y_{r_q}) \geq \lim_q r_q = \alpha.$$

Thus $\min_x p(x, y_*) \geq \alpha$. Together with (S4.17), this proves (S4.16) and shows that the right-hand maximum is attained at y_* . $\square$

Lemma A.15 (Compact coefficient hull and exact finite barycenters). *Let I be a nonempty compact space and let $a : I \rightarrow \mathbb{R}^D$ be continuous. Then*

$$\mathcal{C} := \text{conv}(a(I))$$

is compact and convex. Moreover, every $c \in \mathcal{C}$ has an exact representation

$$c = \sum_{j=1}^r \lambda_j a(i_j), \quad 1 \leq r \leq D+1, \quad \lambda_j \geq 0, \quad \sum_{j=1}^r \lambda_j = 1, \quad (\text{S4.22})$$

with $i_j \in I$. Consequently, for every $K \in \mathbb{R}^D$,

$$\langle K, c \rangle = \sum_{j=1}^r \lambda_j \langle K, a(i_j) \rangle \quad (\text{S4.23})$$

exactly.

Proof. Convexity is immediate from the definition. For the support bound, start from any finite convex representation $c = \sum_{j=1}^q \lambda_j a(i_j)$, which exists by the definition of convex hull, and first remove every zero-weight term. Thus all remaining $\lambda_j > 0$. If $q > D + 1$, the points $a(i_1), \dots, a(i_q)$ are affinely dependent. Hence there are scalars θ_j , not all zero, such that

$$\sum_{j=1}^q \theta_j = 0, \quad \sum_{j=1}^q \theta_j a(i_j) = 0. \quad (\text{S4.24})$$

After changing all signs if needed, at least one $\theta_j > 0$. Put

$$\rho := \min_{j: \theta_j > 0} \frac{\lambda_j}{\theta_j}.$$

Then $\lambda'_j = \lambda_j - \rho \theta_j$ are nonnegative, sum to one, represent the same c by (S4.24), and at least one formerly positive coefficient becomes zero. Removing zero coefficients and iterating gives (S4.22). This is the affine-dependence proof of Caratheodory's theorem.

The representation bound also proves compactness. Pad every representation to exactly $D + 1$ terms using zero weights. Then $\mathcal{C}$ is the image of the compact set

$$\Delta([D + 1]) \times I^{D+1}$$

under the continuous map $(\lambda, i_1, \dots, i_{D+1}) \mapsto \sum_j \lambda_j a(i_j)$. Thus $\mathcal{C}$ is compact. Equation (S4.23) follows from bilinearity of the Euclidean pairing and introduces no approximation. $\square$

Proposition A.4 (Finite public hard prior for source-private one-block kernels). *Let $b_*, d_* > 0$ and $N_* \geq 2$ be the constants supplied by Proposition A.3. Under that proposition and Lemmas A.11–A.15, for every pair of integers N, M satisfying (S4.H), there is a probability law $\mu_{N,M}$ supported on at most*

$$D + 1 = (2N)^M 2^N + 1 \quad (\text{S4.25})$$

pairs in $[N + 1] \times \Delta([N])$ such that every randomized unrestricted $(0.1, d_/(M^2 \log M))$ -DP one-block learner $B : ([N] \times \{0, 1\})^M \rightarrow \{0, 1\}^{[N]}$ satisfies*

$$\mathbb{E}_{(t,Q) \sim \mu_{N,M}} \mathbb{E}_{S \sim (Q^{\tau_t})^M, g \sim B(S)} R_Q(g, \tau_t) > \eta = 2^{-8}.$$

The prior is selected independently of the learner and may be revealed to it.

Proof. Use the coefficient map in (S4.11) and define

$$\mathcal{A}_{N,M} := a_{N,M}(\mathcal{I}_N), \quad \mathcal{C}_{N,M} := \text{conv}(\mathcal{A}_{N,M}).$$

Lemma A.15 makes $\mathcal{C}_{N,M}$ a nonempty compact convex subset of $\mathbb{R}^D$. For every fixed K , linearity of the pairing gives

$$\max_{c \in \mathcal{C}_{N,M}} \langle K, c \rangle = \max_{a \in \mathcal{A}_{N,M}} \langle K, a \rangle = f_{N,M}(K). \quad (\text{S4.26})$$

Indeed, a convex combination cannot exceed the largest paired coefficient, and $\mathcal{A}_{N,M} \subseteq \mathcal{C}_{N,M}$ gives the reverse inequality.

Apply Lemma A.14 with

$$X = \mathcal{K}_{N,M}, \quad Y = \mathcal{C}_{N,M}, \quad p(K, c) = \langle K, c \rangle.$$

The first set is nonempty compact convex by Lemma A.11; the second is nonempty compact convex by Lemma A.15; and the payoff is continuous bilinear. Thus all hypotheses of the separation-based minimax theorem are discharged. Equations (S4.8), (S4.15), and (S4.26) yield the exact quantifier swap

$$\begin{aligned} v_{N,M} &= \min_{K \in \mathcal{K}_{N,M}} \max_{c \in \mathcal{C}_{N,M}} \langle K, c \rangle \\ &= \max_{c \in \mathcal{C}_{N,M}} \min_{K \in \mathcal{K}_{N,M}} \langle K, c \rangle > \eta. \end{aligned} \quad (\text{S4.27})$$

Let $c_* \in \mathcal{C}_{N,M}$ attain the maximum in (S4.27). By Lemma A.15, there are $r \leq D + 1$, pairs $(t_j, Q_j) \in \mathcal{I}_N$, and nonnegative weights λ_j summing to one such that

$$c_* = \sum_{j=1}^r \lambda_j a_{N,M}(t_j, Q_j). \quad (\text{S4.28})$$

Remove zero weights and merge repeated pairs, and define the finite law

$$\mu_{N,M} := \sum_{j=1}^r \lambda_j \delta_{(t_j, Q_j)}. \quad (\text{S4.29})$$

For every $K \in \mathcal{K}_{N,M}$, equations (S4.12), (S4.27), and (S4.28) give

$$\begin{aligned} \mathbb{E}_{(t,Q) \sim \mu_{N,M}} \ell_{N,M}(K; t, Q) &= \sum_{j=1}^r \lambda_j \langle K, a_{N,M}(t_j, Q_j) \rangle \\ &= \langle K, c_* \rangle \\ &\geq \min_{K' \in \mathcal{K}_{N,M}} \langle K', c_* \rangle \\ &= v_{N,M} > \eta. \end{aligned} \quad (\text{S4.30})$$

The first equality is the exact finite-prior expectation, not an approximation to a continuum. Lemma A.11 identifies every source-private randomized unrestricted B with one such K , so (S4.30) is precisely the hard-prior inequality in the proposition and in (S4.HP).

The construction of $\mu_{N,M}$ uses the whole fixed strategy set $\mathcal{K}_{N,M}$, not any selected learner. Once (S4.29) is fixed and made public, a learner whose program refers to (S4.29) still induces some kernel $K \in \mathcal{K}_{N,M}$ if it is source-private; (S4.30) was proved for every such kernel after the prior was fixed. Hence prior awareness does not change the quantifier order.

Finally, no interiority was used. The compact instance space contains $t = 1, N + 1$, every boundary distribution, and every point-mass Q . Equations (S4.9)–(S4.12) remain polynomial and exact when some $Q(x) = 0$. Different thresholds that agree on the support of a point mass may give duplicate coefficient vectors, but duplicates neither invalidate the convex hull nor change (S4.30). $\square$

The compact-kernel and separation arguments culminate in Proposition A.4: one finite law is chosen before the universal private-kernel quantifier, and it gives a strict prior-averaged loss gap for every such kernel.

A.5 Candidate and simulator calibration

Fix the constants from Proposition A.2 and set

$$a = \frac{b_*}{16}, \quad c_\delta = d_*, \quad \varepsilon_0 = 0.1, \quad \alpha_0 = \beta_0 = 2^{-13}. \quad (\text{S5.C1})$$

For local formulas write $L_N = \log_2^* N$ and $M = m_{n,k}$.

Lemma A.16 (Universal calibration and fixed log-star threshold). *Under the universal constants $b_*, d_* > 0$ and $N_* \in \mathbb{Z}_{\geq 2}$ supplied by Proposition A.2, define*

$$q_* := \left\lceil \frac{16}{b_*} \right\rceil, \quad N_0 := \max \{N_*, \text{Tow}_2(q_*)\}, \quad (\text{S5.1})$$

and choose the constants in (S5.C1). Then

$$a, c_\delta, \varepsilon_0 > 0, \quad \alpha_0, \beta_0 \in (0, 1/2), \quad N_0 \in \mathbb{Z}_{\geq 2},$$

all choices are universal and independent of $k, N, n, \varepsilon, \delta$, and every $N \geq N_0$ satisfies

$$N \geq N_*, \quad b_* \log_2^* N \geq 16. \quad (\text{S5.2})$$

In particular, (S5.2) holds at the boundary $N = N_0$.

Proof. Because $b_* > 0$, q_* is an integer at least one and

$$q_* \geq \frac{16}{b_*}. \quad (\text{S5.3})$$

The tower convention gives

$$\log_2^* \text{Tow}_2(q) = q \quad (q \in \mathbb{Z}_{\geq 1}). \quad (\text{S5.4})$$

Indeed, the claim is immediate for $q = 1$, since $\text{Tow}_2(1) = 2$. If it holds at q , then the first base-two logarithm of $\text{Tow}_2(q+1) = 2^{\text{Tow}_2(q)}$ is $\text{Tow}_2(q)$, so exactly one further iteration is needed.

More directly for the comparison needed here, if $N \geq \text{Tow}_2(q_*)$, then for every $0 \leq j < q_*$, monotonicity of the iterated base-two logarithm on the positive range traversed here gives

$$\log_2^{(j)} N \geq \log_2^{(j)} \text{Tow}_2(q_*) > 1.$$

Hence $L_N = \log_2^* N \geq q_*$. Combining this with (S5.3) yields

$$b_* L_N \geq b_* q_* \geq 16. \quad (\text{S5.5})$$

Definition (S5.1) also gives $N \geq N_*$, including when $N = N_0$. The remaining assertions follow from the positivity of b_*, d_* , the exact choices $a = b_*/16$, $c_\delta = d_*$, $\varepsilon_0 = 0.1$, and $2^{-13} \in (0, 1/2)$. Since b_*, d_*, N_* are universal by Proposition A.2, so are q_*, N_0 , and every constant in (S5.C1). $\square$

Lemma A.17 (Strict domination of the exact simulated budget). *Under Assumption 1 and Lemma A.16, if*

$$n < akL_N \quad \text{with} \quad a = \frac{b_*}{16},$$

then the exact setting-defined budget

$$M = \max \left\{ 8, \left\lceil \frac{4n}{k} \right\rceil \right\}$$

is an integer satisfying

$$8 \leq M < b_* L_N. \quad (\text{S5.6})$$

This remains strict when the maximum is attained by the floor-eight branch, when $4n/k$ is an integer, and at $N = N_0$.

Proof. The contradiction hypothesis and $k > 0$ give the strict inequality

$$\frac{4n}{k} < 4aL_N = \frac{1}{4}b_*L_N. \quad (\text{S5.7})$$

For every real x , including integral x , $\lceil x \rceil < x + 1$. By (S5.5), $1 \leq b_*L_N/16$. Therefore

$$\begin{aligned} \left\lceil \frac{4n}{k} \right\rceil &< \frac{4n}{k} + 1 \\ &< \frac{1}{4}b_*L_N + 1 \\ &\leq \frac{5}{16}b_*L_N \\ &< b_*L_N. \end{aligned} \quad (\text{S5.8})$$

The fixed lower branch is also strictly below the source threshold:

$$8 < 16 \leq b_*L_N. \quad (\text{S5.9})$$

Both entries of the maximum are strictly below b_*L_N , so their maximum is strictly below it. Since both entries are integers and the first is 8, $M \in \mathbb{Z}_{\geq 8}$, proving (S5.6).

At the requested smallest tuple $(n, k) = (1, 2)$,

$$\left\lceil \frac{4n}{k} \right\rceil = \lceil 2 \rceil = 2, \quad M = 8.$$

Whenever the local contradiction hypothesis holds, (S5.9) proves the strict hard-regime inequality for this tuple as well. At $N = N_0$, (S5.5) still holds, so no asymptotic or strictly-larger-domain exception is being used. $\square$

Lemma A.18 (Candidate-conjunction preservation and strict source privacy cap). *Under Assumption 1, with $\varepsilon_0 = 0.1$, $0 < c_\delta \leq d_*$, and the exact $M = m_{n,k} \geq 8$, one has simultaneously*

$$0 < \varepsilon \leq 0.1, \quad (\text{S5.10})$$

$$0 < \delta \leq \frac{1}{n \log(n+1)}, \quad (\text{S5.11})$$

$$0 < \delta \leq \frac{c_\delta}{M^2 \log(M+1)} < \frac{d_*}{M^2 \log M}. \quad (\text{S5.12})$$

Thus both original δ -conjuncts are retained, while the second strictly implies the source cap.

Proof. Equation (S5.10) is the candidate ε -condition with the fixed choice $\varepsilon_0 = 0.1$. Since an upper bound by a minimum is an upper bound by each entry, the candidate δ -condition gives (S5.11) and the first inequality in (S5.12) separately. Their denominators are well-defined and positive: $n \geq 1$ gives $\log(n+1) \geq \log 2 > 0$, while $M \geq 8$ gives

$$\log(M+1) > \log M \geq \log 8 > 0. \quad (\text{S5.13})$$

The first inequality in (S5.13) is strict because the natural logarithm is strictly increasing and $M+1 > M$. Taking positive reciprocals reverses that strict comparison. Using $0 < c_\delta \leq d_*$,

$$\frac{c_\delta}{M^2 \log(M+1)} < \frac{c_\delta}{M^2 \log M} \leq \frac{d_*}{M^2 \log M}, \quad (\text{S5.14})$$

which proves the strict part of (S5.12). Notice that (S5.11) was not used to derive (S5.14) and has not been replaced by it.

At the exact floor boundary $M = 8$, (S5.12) reads

$$0 < \delta \leq \frac{c_\delta}{64 \log 9} < \frac{d_*}{64 \log 8}. \quad (\text{S5.15})$$

At $(n, k) = (1, 2)$, where Lemma A.17 computes $M = 8$, the full primitive conjunction is explicitly

$$0 < \delta \leq \frac{1}{\log 2} \quad \text{and} \quad 0 < \delta \leq \frac{c_\delta}{64 \log 9},$$

and (S5.15) supplies the source cap. $\square$

Proposition A.5 (Hard-regime and candidate-parameter certificate). *Under Assumption 1, Proposition A.2, and the local contradiction hypothesis $n < ak \log_2^* N$, the universal choices in (S5.C1) and (S5.1) imply the following complete certificate. The exact $M = m_{n,k}$ satisfies*

$$N \geq N_*, \quad 8 \leq M < b_* \log_2^* N, \quad 0 < \varepsilon \leq 0.1,$$

$$0 < \delta \leq \frac{1}{n \log(n+1)}, \quad 0 < \delta \leq \frac{c_\delta}{M^2 \log(M+1)} < \frac{d_*}{M^2 \log M}.$$

The conclusion includes $N = N_0$, $M = 8$, and, when the contradiction hypothesis is active, $(n, k) = (1, 2)$.

Proof. Lemma A.16 fixes the universal theorem constants once and for all and gives $N \geq N_*$ and $b_* L_N \geq 16$, including at $N = N_0$. Lemma A.17 applies the local contradiction hypothesis to the exact ceiling-and-maximum definition of M , proving $M \in \mathbb{Z}_{\geq 8}$ and the strict inequality $M < b_* L_N$. Lemma A.18 gives $0 < \varepsilon \leq 0.1$, keeps both candidate- δ inequalities, and proves the stronger strict source-cap membership $0 < \delta < d_*/(M^2 \log M)$. The explicit boundary computations (S5.9) and (S5.15), together with the $(n, k) = (1, 2)$ calculation after (S5.9), prove the stated boundary clauses. $\square$

Proposition A.5 therefore retains both candidate- δ conditions while placing the same exact integer M in the strict one-block hard regime under the negated target bound.

A.6 Pointwise PAC expectation control

Lemma A.19 (Every fixed product vector is a realizable instance). *Under the basic setting definitions, fix integers $k \geq 2$, $N \geq 2$ and a deterministic vector*

$$((t_i, Q_i))_{i=1}^k \in ([N+1] \times \Delta([N]))^k.$$

Then

$$P_{\mathbf{Q}}(i, x) := \frac{1}{k} Q_i(x) \quad ((i, x) \in X_{k,N})$$

is a probability distribution on $X_{k,N}$, $\mathbf{t} = (t_1, \dots, t_k) \in [N+1]^k$ defines $\mathbf{c}_{\mathbf{t}} \in C_{k,N}$, and $P_{\mathbf{Q}}^{\mathbf{c}_{\mathbf{t}}}$ is the fixed realizable labeled law generated by $P_{\mathbf{Q}}$ and that concept. These conclusions also hold when any $t_i \in \{1, N+1\}$, when any Q_i is a point mass, or, more generally, when any Q_i lies on a boundary face of $\Delta([N])$.

Proof. For every (i, x) , nonnegativity of $Q_i(x)$ gives $P_{\mathbf{Q}}(i, x) \geq 0$. Since each Q_i is a probability distribution on $[N]$,

$$\sum_{(i,x) \in X_{k,N}} P_{\mathbf{Q}}(i, x) = \frac{1}{k} \sum_{i=1}^k \sum_{x=1}^N Q_i(x) = \frac{1}{k} \sum_{i=1}^k 1 = 1. \quad (\text{S6.1})$$

Thus $P_{\mathbf{Q}}$ is a fixed probability distribution on the exact domain $X_{k,N}$. Each fixed t_i belongs to $[N+1]$, so the displayed class definition gives

$$c_{\mathbf{t}}(i, x) = \tau_{t_i}(x) = \mathbf{1}\{x \geq t_i\}. \quad (\text{S6.2})$$

Thus the function $c_{\mathbf{t}}$, rather than its scalar value in (S6.2), belongs to $C_{k,N}$.

Drawing $Z \sim P_{\mathbf{Q}}$ and assigning the deterministic label $c_{\mathbf{t}}(Z)$ is, by definition, the labeled law $P_{\mathbf{Q}}^{c_{\mathbf{t}}}$. Hence it is realizable by a member of $C_{k,N}$.

No interior-support property was used in (S6.1) or (S6.2). If Q_i is a point mass or has zero coordinates, its coordinates remain nonnegative and sum to one. If $t_i = 1$, then $\tau_{t_i} \equiv 1$ on $[N]$; if $t_i = N+1$, then $\tau_{t_i} \equiv 0$. Both endpoint concepts belong to the displayed class. Nonuniqueness of two thresholds on the support of a degenerate Q_i is irrelevant: the fixed parameter $\mathbf{t}$ still specifies one legal concept and therefore one fixed realizable labeled law. $\square$

Proposition A.6 (Pointwise PAC-to-expectation conversion). *Under Assumption 3, Lemma A.16, and Lemma A.19, let $((t_i, Q_i))_{i=1}^k \in \mathcal{I}_N^k$ be any fixed deterministic vector. For the exact fixed instance $(P_{\mathbf{Q}}, c_{\mathbf{t}})$, exact iid sample size n , and arbitrary randomized output $A(S) \in \mathcal{H}_{k,N}$,*

$$\begin{aligned} \mathbb{E}_{S \sim (P_{\mathbf{Q}}^{c_{\mathbf{t}}})^n} \mathop{\mathbb{E}}_A R_{P_{\mathbf{Q}}}(A(S), c_{\mathbf{t}}) \\ \leq \alpha_0(1 - \beta_0) + \beta_0 \leq \alpha_0 + \beta_0 = 2^{-12}. \end{aligned} \quad (\text{S6.3})$$

The quantifier over the fixed vector precedes both the sample/learner expectation in (S6.3) and any later analysis-side prior average.

Proof. Fix the vector. By Lemma A.19, $P_{\mathbf{Q}}$ is a fixed distribution and $c_{\mathbf{t}} \in C_{k,N}$ realizes its labels. Therefore the universal distribution-free PAC premise applies directly to this same fixed pair. Define the proof-local random variable

$$\mathcal{R} := R_{P_{\mathbf{Q}}}(A(S), c_{\mathbf{t}}),$$

where the only randomness is $S \sim (P_{\mathbf{Q}}^{c_{\mathbf{t}}})^n$ and the internal randomness of A . Population 0-1 risk is a probability of disagreement, so for every possible sample and every possible output hypothesis,

$$0 \leq \mathcal{R} \leq 1. \quad (\text{S6.4})$$

This bound uses no properness or monotonicity: it holds for every $A(S) \in \{0, 1\}^{X_{k,N}}$.

Let

$$G := \{\mathcal{R} \leq \alpha_0\}, \quad p := \Pr(G^c).$$

Assumption 3 gives $p \leq \beta_0$, and $1 - \alpha_0 > 0$ because $\alpha_0 \in (0, 1/2)$. Splitting the expectation over the success and failure events and using (S6.4),

$$\begin{aligned}
\mathbb{E}\mathcal{R} &= \mathbb{E}[\mathcal{R}\mathbf{1}_G] + \mathbb{E}[\mathcal{R}\mathbf{1}_{G^c}] \\
&\leq \alpha_0 \Pr(G) + \Pr(G^c) \\
&= \alpha_0(1-p) + p \\
&= \alpha_0 + (1-\alpha_0)p \\
&\leq \alpha_0 + (1-\alpha_0)\beta_0 \\
&= \alpha_0(1-\beta_0) + \beta_0 \\
&\leq \alpha_0 + \beta_0.
\end{aligned} \tag{S6.5}$$

Lemma A.16 fixes $\alpha_0 = \beta_0 = 2^{-13}$, so

$$\alpha_0 + \beta_0 = 2^{-13} + 2^{-13} = 2^{-12}. \tag{S6.6}$$

Equations (S6.5)–(S6.6) prove (S6.3). Because the vector was fixed before invoking the PAC premise, the proof remains valid for every deterministic vector, including all endpoint-threshold and degenerate-distribution cases covered by Lemma A.19. $\square$

Lemma A.19 applies at every fixed product vector, including endpoints and degenerate block laws. The success/failure split in Proposition A.6 then gives the pointwise expectation ceiling 2^{-12} before any prior averaging.

Assembly of the pointwise and finite-prior consequences. For every deterministic vector

$$((t_i, Q_i))_{i=1}^k \in \mathcal{I}_N^k,$$

first fix that vector and hence fix P_Q and c_t . Lemma A.19 discharges realizability, and Proposition A.6 then gives

$$\mathbb{E}_{S \sim (P_Q^{c_t})^n} R_{P_Q}(A(S), c_t) \leq 2^{-12}. \tag{S6.7}$$

The universal quantifier over the deterministic vector precedes the sample and learner expectation in (S6.7).

Only after this pointwise statement is established do we use the finite law from Proposition A.4. Finite averaging of (S6.7) under $\mu_{N,M}^{\otimes k}$ gives

$$\mathbb{E}_{((t_i, Q_i))_{i=1}^k \sim \mu_{N,M}^{\otimes k}} \left[\mathbb{E}_{S \sim (P_Q^{c_t})^n} R_{P_Q}(A(S), c_t) \right] \leq 2^{-12}. \tag{S6.8}$$

Thus (S6.8) is a consequence of the fixed-vector PAC argument, not an application of the PAC premise to a prior mixture. $\square$

A.7 Block restriction and exact risk

Lemma A.20 (Legality of arbitrary block restriction). *Under the tagged-domain and output-space definitions in Sections 1–2, if $h \in \mathcal{H}_{k,N}$ and $j \in [k]$, then the map $D_j h : [N] \rightarrow \{0, 1\}$ defined by $(D_j h)(x) = h(j, x)$ belongs to the full one-block improper output space $\{0, 1\}^{[N]}$. No monotonicity or threshold-representation condition on h or $D_j h$ is required.*

Proof. Fix arbitrary $h \in \mathcal{H}_{k,N}$, $j \in [k]$, and $x \in [N]$. Then $(j, x) \in X_{k,N}$. Because $h \in \{0, 1\}^{X_{k,N}}$, its value $h(j, x)$ lies in $\{0, 1\}$. Consequently $x \mapsto h(j, x)$ is a binary-valued function on all of $[N]$, hence an element of $\{0, 1\}^{[N]}$.

An improper learner of the threshold targets $\{\tau_t : t \in [N + 1]\}$ may output any element of this full binary-function space. Thus legality does not require $D_j h$ to equal any threshold τ_s , and the argument applies unchanged to constant, oscillating, or otherwise nonmonotone h . $\square$

Proposition A.7 (Exact restriction-risk identity). *Under the primitive threshold and population 0-1 risk definitions and Lemma A.20, if $h \in \mathcal{H}_{k,N}$, $j \in [k]$, and $(\mathbf{t}, \mathbf{Q}) = ((t_i, Q_i))_{i=1}^k$ has $t_i \in [N + 1]$ and each Q_i a probability distribution on $[N]$, then*

$$\Pr_{Y \sim Q_j} [(D_j h)(Y) \neq \tau_{t_j}(Y)] = e_j(h; \mathbf{t}, \mathbf{Q}).$$

This equality remains valid for $t_j = 1$, for $t_j = N + 1$, and for every degenerate or non-full-support Q_j .

Proof. For every $x \in [N]$, the restriction definition and the target definition give the pointwise identity

$$\mathbf{1}\{(D_j h)(x) \neq \tau_{t_j}(x)\} = \mathbf{1}\{h(j, x) \neq \tau_{t_j}(x)\} = \mathbf{1}\{h(j, x) \neq c_{\mathbf{t}}(j, x)\}.$$

Taking the finite Q_j -weighted sum of this pointwise equality yields

$$\begin{aligned} \Pr_{Y \sim Q_j} [(D_j h)(Y) \neq \tau_{t_j}(Y)] &= \sum_{x \in [N]} Q_j(x) \mathbf{1}\{(D_j h)(x) \neq \tau_{t_j}(x)\} \\ &= \sum_{x \in [N]} Q_j(x) \mathbf{1}\{h(j, x) \neq c_{\mathbf{t}}(j, x)\} \\ &= e_j(h; \mathbf{t}, \mathbf{Q}). \end{aligned}$$

There is no inequality or approximation in this calculation. When $t_j = 1$, the target is the all-one function; when $t_j = N + 1$, it is the all-zero function. The same pointwise equality still holds. If Q_j is a point mass, or has zero mass on part of $[N]$, the finite weighted sum simply discards the zero-mass coordinates and remains exact.

Finally, let J be any $[k]$ -valued random variable and let H be any random element of $\mathcal{H}_{k,N}$, with arbitrary dependence between them. Lemma A.20 and the displayed risk identity apply separately to every realization $(J, H) = (j, h)$. Therefore $D_J H$ is almost surely a legal improper one-block output and

$$\sum_{x \in [N]} Q_J(x) \mathbf{1}\{(D_J H)(x) \neq \tau_{t_J}(x)\} = e_J(H; \mathbf{t}, \mathbf{Q})$$

holds pathwise. No independence, learner symmetry, or monotonicity premise is used. $\square$

The restriction is legal for every improper output, and Proposition A.7 identifies its one-block risk exactly, with no projection or residual.

A.8 The ideal hidden-coordinate experiment

Fix a finite public law $\mu_{N,M}$ and an arbitrary randomized learner A . Draw $J \sim \text{Unif}[k]$. Conditional on every $J = j$, draw $\Xi_i = (T_i, Q_i)$ independently from $\mu_{N,M}$ for all $i \in [k]$; then draw independent uniform tags $I_1, \dots, I_n$, features $X_\ell \sim Q_{I_\ell}$ conditionally independently, labels $Y_\ell = \tau_{T_{I_\ell}}(X_\ell)$, and finally $H = A(S^{\text{id}})$ for $S^{\text{id}} = (((I_\ell, X_\ell), Y_\ell))_{\ell=1}^n$. The conclusions to be proved are

$$\Xi \sim \mu_{N,M}^{\otimes k}, \quad J \perp\!\!\!\perp \Xi, \quad (\text{S8.2})$$

$$\mathcal{L}(S^{\text{id}} \mid \Xi) = (P_Q^{c_T})^n, \quad (\text{S8.3})$$

$$\mathbb{E}e_J(H; \mathbf{T}, \mathbf{Q}) = \mathbb{E}R_{P_Q}(H, c_T). \quad (\text{S8.4})$$

Lemma A.21 (Hidden-coordinate instance factorization). *Under Proposition A.4 and the ideal experiment above, if $J \sim \text{Unif}[k]$ and, conditional on each $J = j$, the coordinate instances $\Xi_1, \dots, \Xi_k$ are mutually independent with their common law $\mu_{N,M}$, then*

$$\mathcal{L}(\Xi \mid J = j) = \mu_{N,M}^{\otimes k} \quad \text{for every } j \in [k], \quad (\text{S8.5})$$

and consequently (S8.2) holds. This remains true when $\mu_{N,M}$ has singleton support or repeated coordinate draws occur.

Proof. Let $\mathcal{Z} = \text{supp}(\mu_{N,M})$, which is a nonempty finite set because Proposition A.4 supplies a finitely supported probability law. For every $j \in [k]$ and $\xi = (\xi_1, \dots, \xi_k) \in \mathcal{Z}^k$, the specified conditional independence gives

$$\begin{aligned} \Pr\{J = j, \Xi = \xi\} &= \Pr\{J = j\} \Pr\{\Xi = \xi \mid J = j\} \\ &= \frac{1}{k} \prod_{i=1}^k \mu_{N,M}(\{\xi_i\}). \end{aligned} \quad (\text{S8.6})$$

The product on the right does not depend on j . Summing (S8.6) over j gives

$$\Pr\{\Xi = \xi\} = \prod_{i=1}^k \mu_{N,M}(\{\xi_i\}),$$

which is the product law in (S8.2). Equation (S8.6) then factors as

$$\Pr\{J = j, \Xi = \xi\} = \Pr\{J = j\} \Pr\{\Xi = \xi\},$$

proving independence. If the prior has singleton support, all coordinates are the same deterministic pair almost surely, but the same factorization still proves both the product law and independence from J . Calling one coordinate hidden therefore creates no selection bias. $\square$

Lemma A.22 (Conditional i.i.d. tagged-product sample law). *Under the primitive tagged-product definitions and Lemma A.21, suppose that after (J, Ξ) is generated, the tags $I_1, \dots, I_n$ are drawn independently from $\text{Unif}[k]$, independently of the preceding variables; conditional on $(\Xi, I_1, \dots, I_n)$, the features are independent with $X_\ell \sim Q_{I_\ell}$; and $Y_\ell = \tau_{I_\ell}(X_\ell)$. Then for every realized $\xi = ((t_i, Q_i))_{i=1}^k \in \mathcal{Z}^k$ and every $j \in [k]$,*

$$\mathcal{L}(S^{\text{id}} \mid \Xi = \xi, J = j) = (P_Q^{c_t})^n. \quad (\text{S8.7})$$

In particular, the conditional row kernel is independent of j , and (S8.3) holds.

Proof. Fix $\xi = ((t_i, Q_i))_{i=1}^k$ and j . For any row index ℓ , any $i \in [k]$, $x \in [N]$, and $y \in \{0, 1\}$, the tag, feature, and deterministic-label rules give

$$\begin{aligned} \Pr\{((I_\ell, X_\ell), Y_\ell) = ((i, x), y) \mid \Xi = \xi, J = j\} \\ &= \frac{1}{k} Q_i(x) \mathbf{1}\{y = \tau_{t_i}(x)\} \\ &= P_Q(i, x) \mathbf{1}\{y = c_t(i, x)\} \\ &= P_Q^{c_t}(\{(i, x), y\}). \end{aligned} \quad (\text{S8.8})$$

The first line on the right uses neither j nor any property of the learner. The pairs (I_ℓ, X_ℓ) are mutually independent conditional on Ξ : tags are drawn independently, and the conditional feature draws use independent randomness across rows. Labels are deterministic rowwise functions. Therefore the n labeled rows are conditionally independent and all have the common law in (S8.8), which proves (S8.7).

By Lemma A.21, the conditional law of Ξ itself is the same for every j . Since (S8.7) is also the same for every j , summing over J yields (S8.3). Importantly, (S8.3) is a conditional i.i.d. statement for each fixed product instance. Unconditionally, the rows share the random vector Ξ , so no stronger unconditional i.i.d. assertion is made or needed.

Equation (S8.8) remains valid for $t_i = 1$ or $t_i = N + 1$, when the labels in that block are constant, and when Q_i is a point mass or has zeros. No division by $Q_i(x)$ occurs. It also defines an exact size- n dataset when no sampled tag equals the later-designated hidden coordinate J . $\square$

Lemma A.23 (Persistence of selector independence through an asymmetric learner). *Under Lemmas A.21 and A.22, if A is any randomized map with the domain and codomain in Section 1 and is run on S^{id} with fresh internal randomness, then*

$$J \perp\!\!\!\perp (\Xi, S^{\text{id}}, H). \quad (\text{S8.9})$$

Equivalently, for every $i \in [k]$,

$$\Pr\{J = i \mid \Xi, S^{\text{id}}, H\} = \frac{1}{k} \quad \text{almost surely.} \quad (\text{S8.10})$$

No permutation invariance, tag symmetry, properness, or determinism of A is required.

Proof. Because all involved spaces are finite, represent the randomized learner by its stochastic kernel

$$K_A(h \mid s) := \Pr\{A(s) = h\}, \quad h \in \mathcal{H}_{k,N}.$$

For a fixed instance vector ξ , let $p_\xi(s)$ denote its product-sample mass. This is the mass furnished by Lemma A.22. Equations (S8.6)–(S8.8), followed by the independent learner call, give for every atom with $\xi \in \mathcal{Z}^k$,

$$\begin{aligned} & \Pr\{J = j, \Xi = \xi, S^{\text{id}} = s, H = h\} \\ &= \frac{1}{k} \left(\prod_{i=1}^k \mu_{N,M}(\{\xi_i\}) \right) p_\xi(s) K_A(h \mid s). \end{aligned} \quad (\text{S8.11})$$

Every factor after $1/k$ is independent of j . Summing (S8.11) over j shows that it equals

$$\Pr\{J = j\} \Pr\{\Xi = \xi, S^{\text{id}} = s, H = h\},$$

which proves (S8.9), including zero-mass atoms by omission. The finite conditional-probability formula gives (S8.10) on every positive-mass atom.

The kernel $K_A(h \mid s)$ may depend arbitrarily on every tag and every row of s . Thus (S8.11) does not assert that the law of H is invariant under permutations of tags. It uses only that J is an analysis-side variable not supplied as an argument to A , and that the sample kernel in (S8.7) does not depend on J . The learner may also have the public prior hardwired into its code; this changes K_A but introduces no j -factor. $\square$

Proposition A.8 (Deterministic tagged-product risk decomposition). *Under the primitive product-distribution definition and Proposition A.7, for every fixed $\mathbf{t} \in [N+1]^k$, every vector of probability distributions $\mathbf{Q} = (Q_1, \dots, Q_k)$ on $[N]$, and every arbitrary $h \in \mathcal{H}_{k,N}$,*

$$\begin{aligned} R_{P_{\mathbf{Q}}}(h, c_{\mathbf{t}}) &= \frac{1}{k} \sum_{i=1}^k e_i(h; \mathbf{t}, \mathbf{Q}) \\ &= \frac{1}{k} \sum_{i=1}^k R_{Q_i}(D_i h, \tau_{t_i}). \end{aligned} \tag{S8.12}$$

The equality is deterministic and does not require any distributional symmetry of $\mathbf{Q}$ or any structural property of h .

Proof. Expanding the population risk on the finite tagged domain and using

$$P_{\mathbf{Q}}(i, x) = k^{-1} Q_i(x)$$

gives

$$\begin{aligned} R_{P_{\mathbf{Q}}}(h, c_{\mathbf{t}}) &= \sum_{i=1}^k \sum_{x \in [N]} \frac{1}{k} Q_i(x) \mathbf{1}\{h(i, x) \neq c_{\mathbf{t}}(i, x)\} \\ &= \frac{1}{k} \sum_{i=1}^k \sum_{x \in [N]} Q_i(x) \mathbf{1}\{h(i, x) \neq \tau_{t_i}(x)\} \\ &= \frac{1}{k} \sum_{i=1}^k e_i(h; \mathbf{t}, \mathbf{Q}). \end{aligned} \tag{S8.13}$$

Proposition A.7 identifies the i -th inner sum with $R_{Q_i}(D_i h, \tau_{t_i})$, proving the second equality in (S8.12). The calculation remains exact for unequal block distributions, endpoint thresholds, point masses, and arbitrary nonmonotone or tag-asymmetric hypotheses. $\square$

Lemma A.24 (Finite conditional averaging). *Let $\mathcal{G}$ be a sigma-field, let J take values in $[k]$, and suppose $\Pr(J = i \mid \mathcal{G}) = 1/k$ almost surely for every $i \in [k]$. If $a_1, \dots, a_k$ are integrable $\mathcal{G}$ -measurable random variables, then*

$$\mathbb{E}[a_J \mid \mathcal{G}] = \frac{1}{k} \sum_{i=1}^k a_i \quad \text{almost surely,}$$

and consequently

$$\mathbb{E}a_J = \mathbb{E} \left[\frac{1}{k} \sum_{i=1}^k a_i \right].$$

Proof. The pointwise finite identity $a_J = \sum_{i=1}^k a_i \mathbf{1}\{J = i\}$ and $\mathcal{G}$ -measurability of every a_i give

$$\mathbb{E}[a_J \mid \mathcal{G}] = \sum_{i=1}^k a_i \mathbb{E}[\mathbf{1}\{J = i\} \mid \mathcal{G}] = \sum_{i=1}^k a_i \Pr(J = i \mid \mathcal{G}) = \frac{1}{k} \sum_{i=1}^k a_i.$$

Taking expectations proves the second identity. In the ideal experiment all variables to which this lemma is applied lie in $[0, 1]$, so integrability is automatic. $\square$

Proposition A.9 (Selected-coordinate averaging for arbitrary randomized learners). *Under Propositions A.7 and A.8, and Lemmas A.23 and A.24, for the ideal experiment and every randomized map A specified above,*

$$\mathbb{E}[e_J(H; \mathbf{T}, \mathbf{Q}) \mid \Xi, S^{\text{id}}, H] = R_{P_{\mathbf{Q}}}(H, c_{\mathbf{T}}) \quad \text{almost surely.} \quad (\text{S8.14})$$

Consequently,

$$\begin{aligned} \mathbb{E} R_{Q_J}(D_J H, \tau_{T_J}) &= \mathbb{E} e_J(H; \mathbf{T}, \mathbf{Q}) \\ &= \mathbb{E} R_{P_{\mathbf{Q}}}(H, c_{\mathbf{T}}), \end{aligned} \quad (\text{S8.15})$$

where the expectations retain all prior, sample, selector, and learner randomness.

Proof. Conditional on (Ξ, S^{id}, H) , every quantity $e_i(H; \mathbf{T}, \mathbf{Q})$ is fixed. By (S8.10), the remaining coordinate selector is uniform, so Lemma A.24 gives

$$\begin{aligned} \mathbb{E}[e_J(H; \mathbf{T}, \mathbf{Q}) \mid \Xi, S^{\text{id}}, H] &= \sum_{i=1}^k \frac{1}{k} e_i(H; \mathbf{T}, \mathbf{Q}) \\ &= R_{P_{\mathbf{Q}}}(H, c_{\mathbf{T}}), \end{aligned}$$

where the last equality is the pathwise application of Proposition A.8. Taking expectations and using the second conclusion of Lemma A.24 proves the second equality in (S8.15), which is (S8.4). The first equality in (S8.15) is the pathwise Proposition A.7 at the realized (J, H, T_J, Q_J) .

The proof averages over an independent uniform J after conditioning on the complete realized vector, sample, and output. It never replaces e_J by e_1 , never permutes A , and never assumes the errors $e_1, \dots, e_k$ have the same distribution. Thus deliberate tag asymmetry of A is fully covered. For $k = 2$ and $k = 3$, the conditional sums are exactly $(e_1 + e_2)/2$ and $(e_1 + e_2 + e_3)/3$, respectively. Samples in which the selected tag does not occur require no special conditioning and remain part of the same unconditioned identity. $\square$

Lemmas A.21– A.23 establish the exact stochastic factorization, and Proposition A.9 proves (S8.4) for arbitrary tag-asymmetric learners.

A.9 Usage and overflow

Use

$$M = \max \left\{ 8, \left\lceil \frac{4n}{k} \right\rceil \right\}, \quad (\text{S9.1})$$

and, for the selector and tags above, set

$$U = \sum_{\ell=1}^n \mathbf{1}\{I_{\ell} = J\}. \quad (\text{S9.2})$$

The desired certificate is

$$U \sim \text{Bin}(n, 1/k), \quad \Pr\{U > M\} < 2^{-9}. \quad (\text{S9.3})$$

Lemma A.25 (Exact hidden-tag usage law). *Under the ideal hidden-coordinate tag interface, for every pair of integers $k \geq 2, n \geq 1$, the count U in (S9.2) satisfies, for each $j \in [k]$ and each $u \in \{0, \dots, n\}$,*

$$\Pr\{U = u \mid J = j\} = \binom{n}{u} \left(\frac{1}{k}\right)^u \left(1 - \frac{1}{k}\right)^{n-u}. \quad (\text{S9.4})$$

Consequently $U \sim \text{Bin}(n, 1/k)$ unconditionally. The law includes $U = 0$, and its trial count is exactly n .

Proof. Fix $j \in [k]$. By the tag interface, conditional on $J = j$, the variables

$$B_\ell := \mathbf{1}\{I_\ell = j\}, \quad \ell \in [n],$$

are mutually independent, and

$$\Pr\{B_\ell = 1 \mid J = j\} = \Pr\{I_\ell = j \mid J = j\} = \frac{1}{k}.$$

Thus $U = \sum_{\ell=1}^n B_\ell$. For any fixed set of exactly u indices, the probability that precisely those indicators equal one is $k^{-u}(1-1/k)^{n-u}$. The $\binom{n}{u}$ choices are disjoint, proving (S9.4). The right-hand side does not depend on j . Since $\Pr(J = j) = 1/k$,

$$\begin{aligned} \Pr\{U = u\} &= \sum_{j=1}^k \Pr\{J = j\} \Pr\{U = u \mid J = j\} \\ &= \binom{n}{u} \left(\frac{1}{k}\right)^u \left(1 - \frac{1}{k}\right)^{n-u}. \end{aligned}$$

This is the asserted binomial mass function. In particular, no random block instance, feature, label, learner output, or stopping rule changes the law. $\square$

Lemma A.26 (Falling-factorial binomial tail bound). *Let $V \sim \text{Bin}(n, p)$, where $n \geq 1$ and $p \in [0, 1]$. For every integer $v \geq 0$, set $(v)_0 := 1$; for every integer $r \geq 1$, write*

$$(v)_r := v(v-1) \cdots (v-r+1),$$

with $(v)_r = 0$ when $r > v$. Then, for every integer $r \geq 1$,

$$\mathbb{E}(V)_r = (n)_r p^r \quad (\text{S9.5})$$

and

$$\Pr\{V \geq r\} \leq \frac{(n)_r p^r}{r!} \leq \frac{(np)^r}{r!}. \quad (\text{S9.6})$$

Equations (S9.5)–(S9.6) remain valid when $r > n$, in which case both the event and $(n)_r$ vanish.

Proof. Represent $V = \sum_{\ell=1}^n Z_\ell$ for independent Bernoulli(p) variables. For every outcome with $V = v$, the number of ordered r -tuples of pairwise distinct successful trial indices is exactly $(v)_r$. Hence the pointwise identity

$$(V)_r = \sum_{\substack{(\ell_1, \dots, \ell_r) \in [n]^r \\ \ell_1, \dots, \ell_r \text{ pairwise distinct}}} \prod_{s=1}^r Z_{\ell_s} \quad (\text{S9.7})$$

holds, with an empty sum if $r > n$. There are $(n)_r$ ordered tuples, and independence gives expectation p^r for every product in (S9.7), proving (S9.5).

If $V \geq r$, then

$$(V)_r = r! \binom{V}{r} \geq r!,$$

whereas if $V < r$, both $\mathbf{1}\{V \geq r\}$ and $(V)_r$ are zero. Therefore

$$\mathbf{1}\{V \geq r\} \leq \frac{(V)_r}{r!}$$

pointwise. Taking expectations and using (S9.5) gives the first inequality in (S9.6). Finally, each of the r nonnegative factors of $(n)_r$ is at most n when $r \leq n$, so $(n)_r \leq n^r$. When $r > n$, the left side is zero. This proves the second inequality in every case. $\square$

Lemma A.27 (Floor-eight small-mean bound). *Under Lemmas A.25 and A.26, define*

$$\lambda := \frac{n}{k}. \tag{S9.8}$$

If $\lambda \leq 2$, then the exact budget (S9.1) satisfies $M = 8$ and

$$\Pr\{U > M\} \leq \frac{\lambda^9}{9!} \leq \frac{2^9}{9!} < 2^{-9}. \tag{S9.9}$$

If, more specifically, $n < k$, then $M = 8$ and

$$\Pr\{U > M\} < \frac{1}{9!} < 2^{-9}. \tag{S9.10}$$

Thus the statement includes the exact boundary $\lambda = 2$, every occurrence of $M = 8$, and the case $n < k$.

Proof. If $\lambda \leq 2$, then $4\lambda \leq 8$, so

$$\lceil 4n/k \rceil = \lceil 4\lambda \rceil \leq 8.$$

Equation (S9.1) therefore gives $M = 8$ exactly. Since U is integer,

$$\{U > M\} = \{U > 8\} = \{U \geq 9\}. \tag{S9.11}$$

Apply Lemma A.26 to the exact law in Lemma A.25 with $p = 1/k$ and $r = 9$. Because $np = n/k = \lambda$,

$$\Pr\{U > M\} \leq \frac{\lambda^9}{9!} \leq \frac{2^9}{9!}. \tag{S9.12}$$

The final comparison in (S9.9) is strict and completely numerical:

$$9! = 362880 > 262144 = 2^{18}. \tag{S9.13}$$

Dividing (S9.13) by the positive number $2^9 9!$ gives $2^9/9! < 2^{-9}$, completing (S9.9). If $n < 9$, the event in (S9.11) is actually empty because $U \leq n$, and the convention in Lemma A.26 records this as $(n)_9 = 0$.

If $n < k$, then $0 < \lambda < 1$ and $0 < 4\lambda < 4$, whence $\lceil 4\lambda \rceil \leq 4$ and $M = 8$. The first inequality of (S9.9) is then strict:

$$\Pr\{U > M\} \leq \frac{\lambda^9}{9!} < \frac{1}{9!}.$$

Since $9! = 362880 > 512 = 2^9$, one has $1/9! < 2^{-9}$, proving (S9.10). Conversely, if $M = 8$, then (S9.1) implies $\lceil 4\lambda \rceil \leq 8$, hence $4\lambda \leq 8$ and $\lambda \leq 2$; thus every possible $M = 8$ case is indeed contained in this lemma. $\square$

Lemma A.28 (Monotone factorial envelope). *For every integer $j \geq 9$, define*

$$a_j := \frac{(j/4)^{j+1}}{(j+1)!}. \quad (\text{S9.14})$$

Then $(a_j)_{j \geq 9}$ is strictly decreasing and

$$a_j \leq a_9 = \frac{(9/4)^{10}}{10!} < 2^{-10}. \quad (\text{S9.15})$$

Proof. Direct cancellation gives, for every integer $j \geq 9$,

$$\frac{a_{j+1}}{a_j} = \frac{j+1}{4(j+2)} \left(1 + \frac{1}{j}\right)^{j+1}. \quad (\text{S9.16})$$

We bound the power in (S9.16) without importing an asymptotic exponential estimate. The binomial theorem gives

$$\left(1 + \frac{1}{j}\right)^j = \sum_{r=0}^j \binom{j}{r} j^{-r} = \sum_{r=0}^j \frac{(j)_r}{r! j^r} \leq \sum_{r=0}^j \frac{1}{r!} < 3. \quad (\text{S9.17})$$

For the strict last inequality, $r! \geq 2^{r-1}$ for every $r \geq 2$, with strict inequality for every $r \geq 3$, so

$$\sum_{r=2}^{\infty} \frac{1}{r!} < \sum_{r=2}^{\infty} \frac{1}{2^{r-1}} = 1.$$

Adding the $r = 0, 1$ terms proves (S9.17). Since $j \geq 9$,

$$\left(1 + \frac{1}{j}\right)^{j+1} < 3 \left(1 + \frac{1}{9}\right) = \frac{10}{3} < 4. \quad (\text{S9.18})$$

Substituting (S9.18) into (S9.16) yields the strict ratio bound

$$\frac{a_{j+1}}{a_j} < \frac{j+1}{j+2} < 1. \quad (\text{S9.19})$$

Thus the sequence is strictly decreasing and $a_j \leq a_9$. It remains to verify the endpoint constant. Since $4^{10} = 2^{20}$, the inequality $a_9 < 2^{-10}$ is equivalent to

$$9^{10} < 2^{10} 10!.$$

Both integers are explicit:

$$9^{10} = 3486784401 < 3715891200 = 2^{10} 10!. \quad (\text{S9.20})$$

Equations (S9.19)–(S9.20) prove (S9.15). $\square$

Lemma A.29 (Ceiling-controlled large-mean bound). *Under Lemmas A.25, A.26, and A.28, if $\lambda = n/k > 2$ and*

$$j := \lceil 4\lambda \rceil, \quad (\text{S9.21})$$

then j is an integer satisfying $j \geq 9$, the exact budget is $M = j$, and

$$\Pr\{U > M\} \leq \frac{\lambda^{j+1}}{(j+1)!} \leq \frac{(j/4)^{j+1}}{(j+1)!} \leq \frac{(9/4)^{10}}{10!} < 2^{-10} < 2^{-9}. \quad (\text{S9.22})$$

The conclusion remains valid when $j+1 > n$, when the overflow event is empty.

Proof. The strict inequality $\lambda > 2$ gives $4\lambda > 8$. Therefore its integer ceiling obeys

$$j = \lceil 4\lambda \rceil \geq 9, \quad (\text{S9.23})$$

so (S9.1) gives $M = \max\{8, j\} = j$. Because U is integer,

$$\{U > M\} = \{U \geq j+1\}. \quad (\text{S9.24})$$

Apply Lemma A.26 with $p = 1/k$ and the integer order $r = j+1$. Together with Lemma A.25, this gives

$$\Pr\{U > M\} \leq \frac{(n/k)^{j+1}}{(j+1)!} = \frac{\lambda^{j+1}}{(j+1)!}. \quad (\text{S9.25})$$

If $j+1 > n$, the sharper first expression in (S9.6) has $(n)_{j+1} = 0$, so (S9.24) is empty and (S9.25) still holds. No unstated assumption $j+1 \leq n$ is needed.

The ceiling definition (S9.21) also gives $j \geq 4\lambda$, hence $\lambda \leq j/4$. Raising nonnegative quantities to the positive integer power $j+1$ and dividing by $(j+1)! > 0$ proves the second inequality in (S9.22). Lemma A.28 supplies the next two inequalities, and $2^{-10} < 2^{-9}$ follows from $2^{-10} = \frac{1}{2}2^{-9}$. This proves the entire strict chain (S9.22). $\square$

Proposition A.10 (Exact finite-budget overflow certificate). *Under the basic-setting ranges $k \geq 2, n \geq 1$, the exact budget (S9.1), and the ideal hidden-coordinate tag interface, the usage count (S9.2) satisfies (S9.3). More explicitly:*

1. If $n/k \leq 2$, then $M = 8$ and (S9.9) gives the strict tail bound.
2. If $n/k > 2$, then $M = \lceil 4n/k \rceil \geq 9$ and (S9.22) gives the stronger tail bound $\Pr(U > M) < 2^{-10}$.
3. If $k = 2$ or $k = 3$, then $M \geq n$ and $\Pr(U > M) = 0$.
4. If $n < k$, then $M = 8$ and the strict chain (S9.10) holds.
5. If $M = 8$, necessarily $n/k \leq 2$, so the first branch applies.

No contradiction-regime, hard-prior, privacy, PAC, or learner condition is needed for any clause.

Proof. Lemma A.25 proves the first assertion in (S9.3). Exactly one of $n/k \leq 2$ and $n/k > 2$ holds. In the first case, Lemma A.27 proves $\Pr(U > M) < 2^{-9}$; in the second, Lemma A.29 proves the stronger $\Pr(U > M) < 2^{-10} < 2^{-9}$. This proves the second assertion in (S9.3) for all k, n .

For the requested small-tag boundaries, if $k \in \{2, 3\}$, then $4/k \geq 1$, so

$$\left\lceil \frac{4n}{k} \right\rceil \geq n \quad \text{and hence} \quad M \geq n. \quad (\text{S9.26})$$

The count (S9.2) always obeys $0 \leq U \leq n$. Thus $U > M$ is impossible and its probability is zero. This direct argument covers every n , including values for which $M = 8$ and values for which the ceiling branch exceeds eight.

The $n < k$ and $M = 8$ clauses, including all ceiling inequalities and their strict constants, were proved in Lemma A.27. Finally, $U = 0$ never belongs to the overflow event because $M \geq 8$; it remains an ordinary atom of the exact binomial law rather than a conditioned-away case. $\square$

The exhaustive small- and large-mean estimates in Proposition A.10 prove (S9.3), including zero overflow for $k = 2, 3$.

A.10 The executable one-use simulator

Fix a candidate and, locally for contradiction, suppose

$$n < ak \log_2^* N. \quad (\text{S10.2})$$

With $M = m_{n,k}$, Proposition A.4 provides the finite public law

$$\mu_{N,M} \quad \text{on} \quad \mathcal{I}_N. \quad (\text{S10.3})$$

For an arbitrary $A : \mathcal{S}_n \rightarrow \mathcal{H}_{k,N}$, the simulator to be constructed is

$$B_{\mu_{N,M},A} : \mathcal{Z}_M \longrightarrow \mathcal{G}_N. \quad (\text{S10.5})$$

Lemma A.30 (Input-independent public preprocessing). *Under Assumption 1, the contradiction hypothesis (S10.2), Propositions A.4, A.5, and A.10, the following ordered procedure defines a preprocessing seed Ω_{pre} before any input coordinate is evaluated:*

1. Draw $J \sim \text{Unif}[k]$.
2. Draw $I_1, \dots, I_n$ independently and uniformly from $[k]$, independently of J .
3. Conditional on J , independently draw

$$\Theta_i = (T_i, Q_i) \sim \mu_{N,M}, \quad i \in [k] \setminus \{J\};$$

4. Conditional on $J, I_{1:n}, (\Theta_i)_{i \neq J}$, independently draw

$$X_r^\circ \sim Q_{I_r} \quad \text{for every } r \in [n] \text{ with } I_r \neq J,$$

and set

$$Y_r^\circ := \tau_{T_{I_r}}(X_r^\circ).$$

The seed consists of precisely these realized objects. Its distribution depends only on the public tuple $(k, N, n, M, \mu_{N,M})$, is independent of every $z \in \mathcal{Z}_M$, and contains neither a hidden instance (t, Q) , a block-instance draw Θ_J , nor a synthetic feature or label at a position whose tag equals J . Conditional on the realized selector, tags, and nonhidden instances, the nonhidden labeled rows

$$((I_r, X_r^\circ), Y_r^\circ), \quad r : I_r \neq J, \quad (\text{S10.6})$$

are mutually independent, and each row with tag $i \neq J$ has labeled law $Q_i^{\tau_{T_i}}$ after its tag is suppressed.

Proof. Proposition A.5 discharges the exact N, M conditions for Proposition A.4, so the finite public law $\mu_{N,M}$ exists before the present simulator is defined. The first two draws use the selector/tag product kernel certified by Proposition A.10. Once $J = j$ is fixed, the finite product law

$$\bigotimes_{i \in [k] \setminus \{j\}} \mu_{N,M}$$

defines all nonhidden instances. Each realized Q_i is a probability distribution on the finite set $[N]$, so the further finite conditional product

$$\bigotimes_{\substack{r \in [n] \\ I_r \neq J}} Q_{I_r} \quad (\text{S10.7})$$

defines the nonhidden feature draws. For any fixed values $(x_r)_{r: I_r \neq J}$, (S10.7) gives the exact conditional mass

$$\prod_{\substack{r \in [n] \\ I_r \neq J}} Q_{I_r}(x_r), \quad (\text{S10.8})$$

which proves conditional mutual independence and the stated marginal kernels. Applying the deterministic threshold label $\tau_{T_{I_r}}$ gives exactly the labeled law in the lemma.

All four randomization stages are completed without referring to an input argument. In particular, the sampling code knows the public distribution $\mu_{N,M}$ and its own realized nonhidden pairs (T_i, Q_i) , but it neither draws nor receives (T_J, Q_J) . A later hidden pair (t, Q) is therefore not an input to this procedure. Endpoint draws $T_i = 1$ and $T_i = N + 1$ are executable: their deterministic labels are respectively all one and all zero. Point-mass Q_i 's are also valid factors in (S10.7).

The internal randomness used by A is intentionally not a component of Ω_{pre} . It remains the fresh internal randomization of the one call to the kernel A , if that call is reached. Thus every random choice belonging to preprocessing is fixed before input access, without converting randomized A into a deterministic map. $\square$

Proposition A.11 (Exact hidden-blind size- n row construction). *Under Lemma A.30, fix any realization ω of Ω_{pre} and any arbitrary labeled input*

$$z = (z_1, \dots, z_M) \in \mathbf{Z}_M, \quad z_\ell = (x_\ell, y_\ell). \quad (\text{S10.9})$$

Define, using only the selector and tags in ω ,

$$L_r := \sum_{s=1}^r \mathbf{1}\{I_s = J\}, \quad U := L_n. \quad (\text{S10.10})$$

The value of U is computed before any z_ℓ is evaluated. If $U > M$, the row-construction procedure stops immediately and evaluates no input coordinate. If $U \leq M$, it defines an ordered size- n labeled product dataset

$$\mathcal{P}_\omega(z) = (\tilde{z}_1, \dots, \tilde{z}_n) \in \mathbf{S}_n \quad (\text{S10.11})$$

by the exact rule

$$\tilde{z}_r = \begin{cases} ((J, x_{L_r}), y_{L_r}), & I_r = J, \\ ((I_r, X_r^\circ), Y_r^\circ), & I_r \neq J. \end{cases} \quad (\text{S10.12})$$

Thus, on nonoverflow, the ℓ -th input row is used exactly at the ℓ -th hidden-tag occurrence for every $\ell \in [U]$, every nonhidden row is the pre-drawn row (S10.6), and the rule requires no knowledge of a hidden t or Q . It is valid for every binary label pattern in (S10.9), including patterns inconsistent with every threshold.

Proof. The seed is drawn before the input is accessed by Lemma A.30. Both L_r and U are functions only of the already drawn $J, I_1, \dots, I_n$. Consequently the comparison $U > M$ can be resolved before evaluating any coordinate of (S10.9). On overflow, the procedure takes no later row-construction action, so it never attempts to access a nonexistent $(M + 1)$ -st record.

Suppose $U \leq M$. For each position r with $I_r = J$, (S10.10) gives

$$1 \leq L_r \leq U \leq M. \quad (\text{S10.13})$$

Hence z_{L_r} is an available input record. Since $x_{L_r} \in [N]$, $J \in [k]$, and $y_{L_r} \in \{0, 1\}$, the first line of (S10.12) lies in $X_{k,N} \times \{0, 1\}$, regardless of whether $y_{L_r} = \tau_t(x_{L_r})$ for any t .

For each position with $I_r \neq J$, the seed already contains $X_r^\circ \in [N]$ and $Y_r^\circ = \tau_{T_{I_r}}(X_r^\circ) \in \{0, 1\}$, so the second line of (S10.12) has the same required type. There is one and only one applicable line of (S10.12) for every $r \in [n]$. Thus (S10.11) has exactly n ordered labeled rows and belongs to the full domain S_n of A .

If r is the ℓ -th occurrence of tag J , then by definition $L_r = \ell$, so the first line of (S10.12) uses z_ℓ , including its raw label. This is why no hidden threshold t is needed. The feature distribution Q of a later hidden input law is likewise unnecessary: the procedure takes hidden features directly from the input records. Nonhidden features and labels use only the simulator's publicly drawn instances, whose exact conditional kernels were proved in Lemma A.30. $\square$

Lemma A.31 (Ordered one-use input incidence). *Under Proposition A.11, for every fixed preprocessing seed ω and every arbitrary input $z \in Z_M$, the following statements hold pathwise.*

- If $U > M$, no coordinate of z is evaluated.
- If $U \leq M$, let

$$r_\ell := \min \left\{ r \in [n] : \sum_{s=1}^r \mathbf{1}\{I_s = J\} = \ell \right\}, \quad \ell = 1, \dots, U. \quad (\text{S10.14})$$

Then

$$1 \leq r_1 < \dots < r_U \leq n, \quad \tilde{z}_{r_\ell} = ((J, x_\ell), y_\ell). \quad (\text{S10.15})$$

No row of $\mathcal{P}_\omega(z)$ other than row r_ℓ depends on input coordinate z_ℓ ; coordinates $z_{U+1}, \dots, z_M$ are not evaluated at all.

In particular, each input record has at most one image row. If $U = 0$, the input-access set is empty and all n rows are nonhidden pre-drawn rows. If $U = M$, every input record has exactly one image row. The statement is about row incidence only and makes no differential-privacy claim.

Proof. On overflow, the first bullet is part of the control flow proved in Proposition A.11. Now fix $U \leq M$. The running count L_r in (S10.10) starts at zero, is nondecreasing, and increases by exactly one precisely at the positions with $I_r = J$. Because its terminal value is U , it hits each integer $\ell = 1, \dots, U$ at a unique first position r_ℓ , and those positions are strictly increasing. At r_ℓ , one has $I_{r_\ell} = J$ and $L_{r_\ell} = \ell$, so the first line of (S10.12) gives the identity in (S10.15).

At a hidden position r , the only input coordinate named by (S10.12) is z_{L_r} . The map from hidden positions to positive running counts is the bijection $r_\ell \mapsto \ell$. At a nonhidden position, the

second line of (S10.12) is already fixed by ω and names no input coordinate. Therefore z_ℓ , for $\ell \leq U$, occurs in exactly row r_ℓ , and every $\ell > U$ occurs in no row. This proves the one-use incidence claim for arbitrary features and labels, without any realizability premise. $\square$

Proposition A.12 (Total executable improper one-block simulator). *Under Assumption 1, the contradiction hypothesis (S10.2), Propositions A.4, A.5, and A.10, Lemma A.20, and Lemma A.30, Proposition A.11, and Lemma A.31, the following algorithm defines a total randomized kernel*

$$B_{\mu_{N,M},A} : Z_M \longrightarrow \mathbf{G}_N$$

for every arbitrary randomized map $A : S_n \rightarrow \mathcal{H}_{k,N}$.

On input $z \in Z_M$:

1. Sample the complete seed Ω_{pre} in the order stated by Lemma A.30.
2. Compute U from $J, I_1, \dots, I_n$, without evaluating z .
3. If $U > M$, return the fixed hypothesis

$$g_0(x) := 0 \quad (x \in [N]) \tag{S10.16}$$

immediately, without evaluating any coordinate of z and without invoking A ;

4. If $U \leq M$, construct the exact $\mathcal{P}_{\Omega_{\text{pre}}}(z) \in S_n$ from (S10.12), draw

$$H \sim A(\mathcal{P}_{\Omega_{\text{pre}}}(z))$$

using A 's own fresh internal randomness, and return $D_J H$.

This simulator has exact input size M , makes at most one call to A and only with exact input size n , uses each input row at most once, and is defined on every arbitrary size- M labeled input. It does not know a hidden t, Q , does not require A to be proper or tag-symmetric, and does not infer privacy or an ideal-law coupling.

Proof. The seed law is a valid input-independent probability kernel by Lemma A.30. On the overflow branch, g_0 is a fixed member of $\mathbf{G}_N$; the control flow reaches (S10.16) before any input dereference or call to A . Thus this branch is total and input-independent even when the supplied input has arbitrary corrupt labels.

On the nonoverflow branch, Proposition A.11 proves that the argument of A is a valid member of its complete domain S_n and has exactly n rows. Therefore the randomized map A returns some $H \in \mathcal{H}_{k,N}$. Since $J \in [k]$, Lemma A.20 gives $D_J H \in \mathbf{G}_N$, regardless of whether H is a threshold, monotone, proper, symmetric, deterministic, or tag-asymmetric.

Equivalently, if $E \subseteq \mathbf{G}_N$ and $A(s, h)$ denotes the output probability assigned by A to $h \in \mathcal{H}_{k,N}$ on $s \in S_n$, then the simulator kernel is the well-defined finite expectation

$$B_{\mu_{N,M},A}(z, E) = \mathbb{E}_{\Omega_{\text{pre}}} \left[\begin{cases} \mathbf{1}\{g_0 \in E\}, & U > M, \\ \sum_{h \in \mathcal{H}_{k,N}} A(\mathcal{P}_{\Omega_{\text{pre}}}(z), h) \mathbf{1}\{D_J h \in E\}, & U \leq M. \end{cases} \right]. \tag{S10.17}$$

Every term is nonnegative, the expression equals one when $E = \mathbf{G}_N$, and it is additive over disjoint events. Because all input and output spaces are finite, (S10.17) is a total randomized kernel on every $z \in Z_M$. Formula (S10.17) is a mathematical description of the procedural branches; on $U > M$ it contains no evaluation of $\mathcal{P}_\omega(z)$ and no A -term.

Lemma A.31 proves the exact access pattern. In particular:

- At $U = 0$, no input row is read, all n rows come from the realized nonhidden instances, and A is called once on that size- n dataset;
- At $1 \leq U \leq M$, precisely the first U input rows are read once each, in their original order, and all trailing rows are ignored;
- At $U = M$, all M rows are used once if that event is possible.
- At $U > M$, no row is read and A is not called.

The same clauses apply to endpoint labels, point-mass nonhidden distributions, and arbitrary nonrealizable input labels. For $k = 2, 3$, Proposition A.10 says the overflow branch has probability zero, but the branch remains defined. No step of this proof uses Assumption 2, compares two input datasets, identifies an ideal sample law, or evaluates population risk. Those are separate downstream obligations. $\square$

The ordered construction is total on arbitrary labels. In particular, Lemma A.31 proves the pathwise incidence property, and Proposition A.12 packages the construction as the kernel in (S10.5).

A.11 Actual-to-ideal coupling

Fix $(t, Q) \in \mathcal{I}_N$ and let

$$Z^{\text{in}} \sim (Q^{\tau_t})^M. \quad (\text{S11.3})$$

The fixed-hidden and prior-averaged targets are, respectively,

$$\mathbb{E}_{\text{id},(t,Q)} R_Q(G^{\text{id}}, \tau_t) \geq \mathbb{E} R_Q(B_{\mu_{N,M},A}(Z^{\text{in}}), \tau_t) - \Pr\{U > M\}, \quad (\text{S11.4a})$$

$$\mathbb{E}_{\text{ideal}} R_{Q_J}(D_J H^{\text{id}}, \tau_{T_J}) \geq \mathbb{E}_{\substack{(T,Q) \sim \mu_{N,M} \\ Z^{\text{in}} \sim (Q^{\tau_T})^M}} R_Q(B_{\mu_{N,M},A}(Z^{\text{in}}), \tau_T) - \Pr\{U > M\}. \quad (\text{S11.4b})$$

Lemma A.32 (Shared realization of a finite randomized learner). *For every randomized map $A : (X_{k,N} \times \{0, 1\})^n \rightarrow \mathcal{H}_{k,N}$, there is a deterministic function*

$$\Phi_A : (X_{k,N} \times \{0, 1\})^n \times (0, 1) \longrightarrow \mathcal{H}_{k,N}$$

such that, for $V \sim \text{Unif}(0, 1)$, $\Phi_A(s, V)$ has exactly the law $A(s)$ for every $s \in (X_{k,N} \times \{0, 1\})^n$. For the same V , equal input datasets give equal outputs pathwise. No properness, determinism, or tag symmetry of A is required.

Proof. Enumerate the finite set $\mathcal{H}_{k,N} = \{h_1, \dots, h_q\}$. For each dataset s , put

$$c_m(s) := \sum_{a=1}^m \Pr\{A(s) = h_a\}, \quad c_0(s) := 0,$$

and let $\Phi_A(s, v) = h_m$ at the first index satisfying $c_m(s) \geq v$. Since $c_q(s) = 1$, this is total, and

$$\Pr\{\Phi_A(s, V) = h_m\} = c_m(s) - c_{m-1}(s) = \Pr\{A(s) = h_m\}.$$

This proves the exact learner marginal. Equal datasets and the same V give the same two arguments to the same deterministic function, proving the pathwise synchronization statement. $\square$

Lemma A.33 (Fixed-hidden common-randomness coupling). *Under Propositions A.10 and A.12 and Lemma A.32, fix any $(t, Q) \in \mathcal{I}_N$ and any arbitrary randomized A . There exists a joint probability kernel carrying*

1. An infinite sequence

$$W_\ell = (X_\ell^\star, Y_\ell^\star), \quad \ell \geq 1, \quad (\text{S11.7})$$

of mutually independent draws from Q^{τ_t} , so $X_\ell^\star \sim Q$ and $Y_\ell^\star = \tau_t(X_\ell^\star)$;

2. Independently of (S11.7), the complete input-independent preprocessing seed of Lemma A.30; and

3. Independently of both, one $V \sim \text{Unif}(0, 1)$.

such that the following statements hold.

- The actual input is

$$Z_\ell^{\text{in}} = W_\ell, \quad \ell = 1, \dots, M, \quad (\text{S11.8})$$

and hence has exactly the law (S11.3), independently of simulator preprocessing.

- Put $\Theta_J = (T_J, Q_J) := (t, Q)$, retain the seed's $\Theta_i = (T_i, Q_i)$ for $i \neq J$, write $\Xi = (\Theta_1, \dots, \Theta_k)$, and define the always-present ideal ordered dataset

$$S^{\text{id}} = (\hat{z}_1, \dots, \hat{z}_n) \quad (\text{S11.9})$$

by

$$\hat{z}_r = \begin{cases} ((J, X_{L_r}^\star), Y_{L_r}^\star), & I_r = J, \\ ((I_r, X_r^\circ), Y_r^\circ), & I_r \neq J, \end{cases} \quad r \in [n]. \quad (\text{S11.10})$$

The infinite sequence makes (S11.10) well defined even when $U > M$.

- There is a deterministic realization Φ_A of the learner kernel for which

$$H^{\text{id}} := \Phi_A(S^{\text{id}}, V), \quad G^{\text{id}} := D_J H^{\text{id}} \quad (\text{S11.11})$$

has the exact ideal call-and-restrict marginal.

- Define the actual output by the simulator control flow:

$$G^{\text{act}} := \begin{cases} g_0, & U > M, \\ D_J \Phi_A(\mathcal{P}_{\Omega_{\text{pre}}}(Z^{\text{in}}), V), & U \leq M. \end{cases} \quad (\text{S11.12})$$

Its marginal is exactly $B_{\mu_{N,M}, A}(Z^{\text{in}})$ with input law (S11.3). In the first line of (S11.12), V , (S11.7), and every other coupling-only variable are ignored: operationally the actual simulator still reads no input and makes no call to A .

Thus (S11.7)–(S11.12) give a coupling of the exact actual and ideal experiments, with every expectation and source of randomness specified.

Proof. The finite law Q^{τ_t} defines the product sequence (S11.7). Only its first M coordinates are exposed as the actual input, so (S11.8) is exactly an iid size- M input rather than a stopped or conditioned sample. By construction this sequence is independent of the preprocessing seed, as required by Lemma A.30.

The running indices satisfy $1 \leq L_r \leq U \leq n$ at every hidden position. Hence the infinite sequence gives every hidden row in (S11.10), including on overflow. Nonhidden rows in (S11.10) are the identical realized rows already present in the preprocessing seed. Therefore (S11.9) is always an ordered member of the full size- n learner domain.

Lemma A.32 makes (S11.11) an exact call to the randomized kernel A , followed by the same restriction used by the simulator. In (S11.12), the nonoverflow branch applies the same realization to the exact row map, while the overflow branch is the exact fixed output from Proposition A.12. Sampling an unused coupling variable does not create an operational learner call. Equations (S11.8) and (S11.12), together with the input-independent seed law and that lemma, prove the actual marginal claim. Equations (S11.9)–(S11.11) define the ideal marginal. $\square$

Lemma A.34 (Identification with the ideal experiment). *Under Lemmas A.21, A.22, and A.23, and under Lemma A.33, fix $j \in [k]$ and a complete vector*

$$\boldsymbol{\theta} = ((t_1, Q_1), \dots, (t_k, Q_k)) \quad (\text{S11.14})$$

with $(t_j, Q_j) = (t, Q)$. Conditional on $J = j$ and on the nonhidden seed instances taking the values in (S11.14), but without conditioning on U , $U \leq M$, or any tag-count event, the ideal dataset (S11.9) satisfies

$$S^{\text{id}} \sim \left(P_Q^{ct}\right)^n. \quad (\text{S11.15})$$

Equivalently, the n coupled ideal global labeled rows are mutually independent, and each has the fixed realizable law that first selects a tag uniformly and then draws its feature from that tag's fixed Q_i . If the nonhidden instances are not fixed, their marginal is only the corresponding mixture of the product laws in (S11.15); this lemma does not call that mixture iid from one deterministic product instance. Moreover, if the hidden pair (t, Q) is first drawn from $\mu_{N,M}$, the complete joint marginal $(J, \Xi, S^{\text{id}}, H^{\text{id}})$ is exactly the joint law characterized by Lemmas A.21–A.23, and

$$\mathbb{E}_{(t,Q) \sim \mu_{N,M}} \mathbb{E}_{\text{id},(t,Q)} R_Q(G^{\text{id}}, \tau_t) = \mathbb{E}_{\text{ideal}} R_{Q_J}(D_J H^{\text{id}}, \tau_{T_J}). \quad (\text{S11.15a})$$

Proof. Fix ordered tags $i_1, \dots, i_n \in [k]$ and features $x_1, \dots, x_n \in [N]$. The tag kernel gives

$$\Pr\{I_1 = i_1, \dots, I_n = i_n \mid J = j\} = k^{-n}. \quad (\text{S11.16})$$

Conditional on this tag vector and (S11.14), enumerate the positions with tag j in increasing order. Formula (S11.10) assigns them the distinct sequence variables $X_1^*, \dots, X_u^*$, where u is the number of such positions. Their joint mass is the product of the corresponding Q_j factors. At every position with tag $i_r \neq j$, Lemma A.30 supplies an independent feature with law Q_{i_r} . Those nonhidden features are conditionally mutually independent and are independent of the separately drawn hidden sequence (S11.7). Therefore

$$\Pr\{\text{the ideal feature at row } r \text{ is } x_r \text{ for all } r \mid J = j, \boldsymbol{\theta}, I_{1:n} = i_{1:n}\} = \prod_{r=1}^n Q_{i_r}(x_r). \quad (\text{S11.17})$$

All labels in (S11.10) are deterministic and equal $\tau_{t_{i_r}}(x_r)$. Multiplying (S11.16) and (S11.17) gives

$$\prod_{r=1}^n \left[\frac{1}{k} Q_{i_r}(x_r) \mathbf{1}\{y_r = \tau_{t_{i_r}}(x_r)\} \right] \quad (\text{S11.18})$$

for every ordered labeled row vector. Expression (S11.18) is exactly the product mass function in (S11.15). This calculation sums over all tag vectors, including those with $U > M$; no overflow conditioning appears.

Now draw the hidden pair $(t, Q) \sim \mu_{N,M}$ independently of the selector before applying the fixed-hidden kernel. Conditional on $J = j$, this pair occupies coordinate j , while Lemma A.30 draws every coordinate $i \neq j$ independently from the same $\mu_{N,M}$. These are precisely the hypotheses of Lemma A.21; hence the full vector has law $\mu_{N,M}^{\otimes k}$ and is independent of J . The row kernel verified in (S11.16)–(S11.18) is precisely the kernel in Lemma A.22. Finally, Lemma A.32 gives the exact same learner kernel A . Thus the complete coupled ideal marginal equals the ideal experiment, and its selected restriction is $D_J H^{\text{id}}$, which proves (S11.15a). Lemma A.23 and Proposition A.9 therefore remain valid for this marginal; their exchangeability and product-risk conclusions are consumed as facts and are not rederived here.

Conditioning (S11.15) further on $U \leq M$ would restrict the tag vectors in (S11.16) and generally destroy the iid tag law. The proof never makes that conditional-iid assertion: later equality on $U \leq M$ is pathwise under the coupling, while (S11.15) is the unconditioned ideal sample law for the fixed block vector. $\square$

Proposition A.13 (Exact row and restricted-output identity off overflow). *Under Lemma A.33, Proposition A.11, and Lemma A.31, on the event*

$$E := \{U \leq M\}, \quad (\text{S11.19})$$

the actual ordered dataset and ideal ordered dataset satisfy, row by row,

$$\mathcal{P}_{\Omega_{\text{pre}}}(Z^{\text{in}})_r = \widehat{z}_r, \quad r = 1, \dots, n. \quad (\text{S11.20})$$

Consequently the shared randomized learner outputs and restrictions in (S11.11)–(S11.12) agree pathwise on E , and

$$R_Q(G^{\text{id}}, \tau_t) = R_Q(G^{\text{act}}, \tau_t) \quad \text{on } E. \quad (\text{S11.21})$$

This includes $U = 0$ and $U = M$. No dataset, output, or risk equality is claimed on $E^c = \{U > M\}$.

Proof. Fix an outcome in E and a row $r \in [n]$. If $I_r = J$, then

$$1 \leq L_r \leq U \leq M. \quad (\text{S11.22})$$

The exact occurrence map in Proposition A.11 uses actual input row L_r . By (S11.8), this row is

$$Z_{L_r}^{\text{in}} = (X_{L_r}^*, Y_{L_r}^*).$$

The hidden-row line of the row rule in Proposition A.11 is therefore

$$((J, X_{L_r}^*), Y_{L_r}^*),$$

which is exactly the first line of the ideal rule (S11.10). If $I_r \neq J$, the nonhidden-row line of Proposition A.11 and the nonhidden-row line of (S11.10) are literally the same realized nonhidden row $((I_r, X_r^\circ), Y_r^\circ)$. This exhaustive row split proves (S11.20), with ordering preserved.

At $U = 0$, there is no hidden row and the second-line identity applies at all n positions; both experiments call A on the identical all- nonhidden dataset. At $U = M$, (S11.22) still holds, and the exact occurrence map may use all and only the first M sequence records. Thus neither endpoint requires a separate conditioning argument or an extra record.

Because (S11.20) gives equal datasets and both calls use the same V , Lemma A.32 gives equal hypotheses even when A is randomized, improper, nonmonotone, or tag-asymmetric. Both sides

then apply the same restriction D_J , so the one-block outputs are equal. Evaluating the identical functions against the same fixed (Q, τ_t) proves (S11.21).

On E^c , Proposition A.12 makes the actual side return g_0 before reading the input or calling A , whereas the ideal side still uses (S11.9)–(S11.11). The proof makes no comparison between those datasets or outputs there. $\square$

Proposition A.14 (Bounded selected-risk transfer). *Under Lemma A.34, Proposition A.13, and Proposition A.10, for every fixed hidden pair (t, Q) and arbitrary randomized A , define*

$$L_{\text{id}} := R_Q(G^{\text{id}}, \tau_t), \quad L_{\text{act}} := R_Q(G^{\text{act}}, \tau_t). \quad (\text{S11.23})$$

Then, on the full coupling space,

$$L_{\text{id}} \geq L_{\text{act}} - \mathbf{1}\{U > M\}. \quad (\text{S11.24})$$

Taking the exact marginals from Lemma A.33 proves (S11.4a). Averaging the hidden pair under $\mu_{N,M}$ and applying Lemma A.34 proves the exact ideal interface (S11.4b). Proposition A.10 supplies the strict residual in (S9.3). If $k = 2$ or $k = 3$, then $U \leq M$ pathwise and the same result has zero residual.

Proof. Population 0-1 risk is a probability, so for every output function,

$$0 \leq L_{\text{id}} \leq 1, \quad 0 \leq L_{\text{act}} \leq 1. \quad (\text{S11.25})$$

On E , Proposition A.13 gives $L_{\text{id}} = L_{\text{act}}$, so (S11.24) holds with a zero indicator. On E^c , (S11.25) gives

$$L_{\text{act}} - 1 \leq 0 \leq L_{\text{id}},$$

which is exactly (S11.24) with indicator one. This overflow argument uses only boundedness. In particular, the actual early-abort output is g_0 , but its risk may be any value in $[0, 1]$: it is zero for the all-zero target, one for the all-one target under any Q , and can take an intermediate value for other thresholds. No equality or favorable abort-risk property is assumed.

Taking expectation of (S11.24) over the joint variables in Lemma A.33 gives

$$\mathbb{E}_{\text{id}, \theta} L_{\text{id}} \geq \mathbb{E}_{\text{act}, \theta} L_{\text{act}} - \mathbb{E} \mathbf{1}\{U > M\}. \quad (\text{S11.26})$$

The actual marginal identity in that lemma identifies the middle term with the simulator expectation displayed in (S11.4a), including all of its preprocessing and learner randomness. The ideal marginal identifies the first term with the ideal selected-risk expectation. Finally,

$$\mathbb{E} \mathbf{1}\{U > M\} = \Pr\{U > M\} < 2^{-9} \quad (\text{S11.27})$$

by Proposition A.10. There is no conditioning-to-expectation conversion and no interchange of a limit, supremum, or integral: the ideal dataset uses only finitely many coordinates of (S11.7), and all other spaces are finite.

Because (S11.26) holds for every fixed (t, Q) , average it over $(t, Q) \sim \mu_{N,M}$. The overflow term is unchanged because its law depends only on $J, I_{1:n}$. Lemma A.34, specifically (S11.15a), identifies the averaged ideal term with the selected-risk quantity in Proposition A.9. The averaged actual term is the right-hand simulator expectation in (S11.4b). This proves (S11.4b) without rederiving or using the equality between ideal selected risk and product population risk.

For $k = 2, 3$, Proposition A.10 gives $M \geq n \geq U$, so (S11.21) holds everywhere and the final term in (S11.26) is exactly zero. For $n < k$ or $M = 8$, the coupling and pointwise inequality are unchanged, while the floor-eight estimate supplies the stated strict residual. $\square$

The shared realization of A synchronizes outputs whenever the ordered datasets agree. Proposition A.14 proves (S11.4a)–(S11.4b) with one and only one overflow charge.

A.12 One-charge privacy of the simulator

Continue under (S10.2). For every $z \simeq z'$ in Z_M and every $E \subseteq G_N$, the target simulator inequalities are

$$B_{\mu_{N,M},A}(z, E) \leq e^\varepsilon B_{\mu_{N,M},A}(z', E) + \delta, \quad (\text{S12.4})$$

$$B_{\mu_{N,M},A}(z, E) \leq e^{0.1} B_{\mu_{N,M},A}(z', E) + \Delta_M, \quad \Delta_M = \frac{d_*}{M^2 \log M}. \quad (\text{S12.6})$$

Lemma A.35 (Deterministic one-row preprocessing map). *Under Assumption 1, the contradiction hypothesis (S10.2), and Proposition A.11 and Lemma A.31, fix any realization ω of the input-independent preprocessing seed and any arbitrary $z, z' \in Z_M$ with $z \simeq z'$. Then exactly one of the following holds.*

1. If $U(\omega) > M$, the preprocessing control flow evaluates no coordinate of either input, so its output branch is identical on z, z' .
2. If $U(\omega) \leq M$, the two exact size- n datasets $\mathcal{P}_\omega(z)$ and $\mathcal{P}_\omega(z')$ are equal or replacement-adjacent. More precisely, if the inputs differ only at row q , then they are equal when $q > U$, while for $q \leq U$ they can differ only at the single global row r_q , the q -th hidden-tag occurrence.

This assertion includes $U = 0$, a changed unused record, a changed used record, simultaneous replacement of the record's feature and label, and arbitrary possibly nonrealizable label patterns. All nonhidden global rows and all hidden rows other than r_q coincide. In particular, the map never produces group-adjacent datasets.

Proof. Proposition A.11 gives

$$U = \sum_{r=1}^n \mathbf{1}\{I_r = J\}, \quad (\text{S12.10})$$

as a function only of $J, I_{1:n}$, all of which belong to ω . Hence the same value of U , and therefore the same overflow branch, is used on z and z' . When $U > M$, Proposition A.11 stops before either input is read. This proves the first branch pathwise, not only in distribution.

Suppose $U \leq M$. Lemma A.31 gives distinct hidden positions

$$1 \leq r_1 < \dots < r_U \leq n \quad (\text{S12.11})$$

depending only on ω , and its exact row rule gives, for $z_\ell = (x_\ell, y_\ell)$,

$$(\mathcal{P}_\omega(z))_{r_\ell} = ((J, x_\ell), y_\ell), \quad \ell = 1, \dots, U. \quad (\text{S12.12})$$

Every other position is a nonhidden row already fixed by ω . The same formulas, with primes, hold for z' , with exactly the same positions r_ℓ and exactly the same nonhidden rows.

If $z = z'$, the synthesized datasets are equal. Otherwise adjacency in Z_M supplies one index $q \in [M]$ such that $z_\ell = z'_\ell$ for every $\ell \neq q$. If $q > U$, Lemma A.31 says row q is unused, and (S12.12) shows equality of every used hidden row; all nonhidden rows also coincide. Thus

$$\mathcal{P}_\omega(z) = \mathcal{P}_\omega(z'). \quad (\text{S12.13})$$

If $q \leq U$, then for every hidden row r_ℓ with $\ell \neq q$, the equality $z_\ell = z'_\ell$ and (S12.12) give equal rows. At the sole remaining hidden position,

$$(\mathcal{P}_\omega(z))_{r_q} = ((J, x_q), y_q), \quad (\mathcal{P}_\omega(z'))_{r_q} = ((J, x'_q), y'_q). \quad (\text{S12.14})$$

Equation (S12.14) is one replacement even if both $x_q \neq x'_q$ and $y_q \neq y'_q$. It is still a legal replacement when either raw label is inconsistent with every threshold because both labels remain elements of $\{0, 1\}$, and Assumption 2 is stated on all labeled datasets. Every nonhidden row is a component of the fixed seed, so no nonhidden row changes. Thus the synthesized datasets differ in at most the one row r_q , proving replacement adjacency and excluding any group-adjacency charge.

Finally, if $U = 0$, there are no positions in (S12.11), every synthesized row is nonhidden and seed-fixed, and (S12.13) holds for every pair of inputs. This also proves the requested null case. $\square$

Proposition A.15 (Fixed-seed privacy kernel with learner coins retained). *Under Assumptions 1 and 2, the contradiction hypothesis (S10.2), Proposition A.12, and Lemma A.35, fix a preprocessing seed ω but do not fix the internal coins of A . Define the all-zero global hypothesis*

$$h_0(i, x) := 0 \quad ((i, x) \in X_{k,N}), \quad (\text{S12.15})$$

and the global-output kernel $K_\omega : Z_M \rightarrow \mathcal{H}_{k,N}$ by

$$K_\omega(z, F) = \begin{cases} \mathbf{1}\{h_0 \in F\}, & U(\omega) > M, \\ \Pr_A[A(\mathcal{P}_\omega(z)) \in F], & U(\omega) \leq M, \end{cases} \quad (\text{S12.16})$$

for every $F \subseteq \mathcal{H}_{k,N}$. Then, for all arbitrary $z \simeq z'$ in Z_M ,

$$K_\omega(z, F) \leq e^\varepsilon K_\omega(z', F) + \delta \quad (F \subseteq \mathcal{H}_{k,N}). \quad (\text{S12.17})$$

On overflow the two output laws coincide exactly. On nonoverflow, (S12.17) uses Assumption 2 at most once. The statement holds for arbitrary randomized A ; its internal coins are precisely the randomness under $\Pr_A$ in (S12.16), not part of the fixed seed.

Proof. The function h_0 belongs to the full output space $\mathcal{H}_{k,N}$, and $D_J h_0 = g_0$ for every J . Thus the first line of (S12.16) is an analytical global-output lift of the simulator's actual fixed overflow output; it does not call A .

If $U(\omega) > M$, the seed fixes the same branch for both inputs, and

$$K_\omega(z, F) = \mathbf{1}\{h_0 \in F\} = K_\omega(z', F). \quad (\text{S12.18})$$

Since $e^\varepsilon \geq 1$ and $\delta \geq 0$, (S12.18) implies (S12.17). This also shows directly that overflow contributes no privacy residual.

Suppose $U(\omega) \leq M$. By Lemma A.35, $\mathcal{P}_\omega(z) = \mathcal{P}_\omega(z')$ or $\mathcal{P}_\omega(z) \simeq \mathcal{P}_\omega(z')$. In the equality case the two laws in (S12.16) coincide, including when $U = 0$ or the changed input row is unused. In the adjacent case, Assumption 2 applied once to the exact size- n pair and the event F gives

$$\begin{aligned} K_\omega(z, F) &= \Pr_A[A(\mathcal{P}_\omega(z)) \in F] \\ &\leq e^\varepsilon \Pr_A[A(\mathcal{P}_\omega(z')) \in F] + \delta \\ &= e^\varepsilon K_\omega(z', F) + \delta. \end{aligned} \quad (\text{S12.19})$$

No coin of A was conditioned on or fixed in (S12.19). Arbitrary feature and label replacements are covered because the premise of (S12.19) is the replacement adjacency proved in Lemma A.35,

not realizability. There is one invocation of Assumption 2, so neither group privacy nor sequential composition appears. $\square$

Lemma A.36 (Common input-independent mixture and restriction preserve one charge). *Under Assumptions 1 and 2, the contradiction hypothesis (S10.2), Lemma A.30 and Proposition A.12, and Proposition A.15, let ν be the law of the input-independent preprocessing seed on its finite seed space Ω_{pre} . For $G \subseteq \Omega_{\text{pre}} \times \mathcal{H}_{k,N}$, define its seed section*

$$G_\omega := \{h \in \mathcal{H}_{k,N} : (\omega, h) \in G\}, \quad (\text{S12.20})$$

and form the joint seed/global-output kernel

$$\hat{K}(z, G) := \sum_{\omega \in \Omega_{\text{pre}}} \nu(\omega) K_\omega(z, G_\omega). \quad (\text{S12.21})$$

Then $\hat{K}$ is (ε, δ) -DP. Applying only after this common mixture the deterministic postprocessing map

$$\Phi(\omega, h) := D_{J(\omega)} h \quad (\text{S12.22})$$

produces exactly the simulator law. Consequently (S12.4) holds with exactly one additive δ .

Proof. The seed space is finite: Lemma A.30 samples finitely many tags, finitely many elements of the finite-support prior, and features from the finite domain $[N]$. For each G , every section G_ω in (S12.20) is therefore a valid event in the global output space. Equation (S12.21) is a total kernel: all summands are nonnegative, its value on $\Omega_{\text{pre}} \times \mathcal{H}_{k,N}$ is $\sum_{\omega} \nu(\omega) K_\omega(z, \mathcal{H}_{k,N}) = 1$, and finite additivity follows by taking disjoint seed sections and using additivity of each K_ω . The seedwise inequality (S12.17) is uniform in ω and in the event, so for arbitrary $z \simeq z'$,

$$\begin{aligned} \hat{K}(z, G) &= \sum_{\omega} \nu(\omega) K_\omega(z, G_\omega) \\ &\leq \sum_{\omega} \nu(\omega) (e^\varepsilon K_\omega(z', G_\omega) + \delta) \\ &= e^\varepsilon \sum_{\omega} \nu(\omega) K_\omega(z', G_\omega) + \delta \sum_{\omega} \nu(\omega) \\ &= e^\varepsilon \hat{K}(z', G) + \delta. \end{aligned} \quad (\text{S12.23})$$

Here Lemma A.30 supplies the same input-independent weights $\nu(\omega)$ for z and z' . Also,

$$\sum_{\omega} \nu(\omega) = 1.$$

Thus (S12.23) first proves exact DP for the common joint mixture. The overflow subset $\{\omega : U(\omega) > M\}$, the $U = 0$ subset, and every nonoverflow subset are determined solely by this common seed. No input-dependent conditioning, normalization, or change of measure occurs.

Now fix $E \subseteq \mathcal{G}_N$ and use the one joint-output event

$$G_E := \Phi^{-1}(E) = \{(\omega, h) : D_{J(\omega)} h \in E\}. \quad (\text{S12.24})$$

On an overflow seed, (S12.15) gives $\Phi(\omega, h_0) = g_0$, so that seed's contribution to $\hat{K}(z, G_E)$ is $\nu(\omega) \mathbf{1}\{g_0 \in E\}$, independently of z . On a nonoverflow seed, its contribution is

$$\nu(\omega) \Pr_A[D_{J(\omega)}A(\mathcal{P}_\omega(z)) \in E], \quad (\text{S12.25})$$

with all internal randomness of A retained. Summing these two exact branches is precisely Proposition A.12, so

$$B_{\mu_{N,M},A}(z, E) = \widehat{K}(z, G_E). \quad (\text{S12.26})$$

Finally apply (S12.23) to $G = G_E$ and use (S12.26) for both inputs. This is the event-preimage proof of postprocessing after the common mixture and gives (S12.4). The additive term in (S12.23) is δ , not δ times the number of seeds, n , k , U , or an overflow factor. $\square$

Proposition A.16 (Exact source-cap membership). *Under Assumptions 1 and 2, the contradiction hypothesis (S10.2), Propositions A.5 and A.12, and Lemma A.36, define the exact source-cap learner set at the present N, M by*

$$\begin{aligned} \mathcal{K}_{N,M}^{\text{src}} := \{ & K : \mathbf{Z}_M \longrightarrow \mathbf{G}_N \text{ a total randomized kernel :} \\ & K(z, E) \leq e^{0.1} K(z', E) + \Delta_M \\ & \text{for all } z \simeq z' \text{ in } \mathbf{Z}_M \text{ and } E \subseteq \mathbf{G}_N \}. \end{aligned} \quad (\text{S12.27})$$

Then

$$B_{\mu_{N,M},A} \in \mathcal{K}_{N,M}^{\text{src}}. \quad (\text{S12.28})$$

More fully, the same parameters satisfy

$$N \geq N_*, \quad 8 \leq M < b_* \log_2^* N, \quad 0 < \varepsilon \leq 0.1, \quad 0 < \delta < \Delta_M,$$

the simulator has exact input size M , full arbitrary one-block output space $\mathbf{G}_N$, and is total on all labeled inputs, and its event probabilities satisfy (S12.6). Thus (S12.28) is exact source-interface membership, not an inference from the phrase “ (ε, δ) -DP” alone.

Proof. Proposition A.12 proves that $B_{\mu_{N,M},A}$ is a total randomized kernel with domain exactly $\mathbf{Z}_M$ and codomain exactly the full improper space $\mathbf{G}_N$. Its totality includes overflow, $U = 0$, arbitrary corrupt input labels, and unused trailing records.

Proposition A.5, applied to the same $N, M, \varepsilon, \delta$, gives $N \geq N_*$, $8 \leq M < b_* \log_2^* N$, $0 < \varepsilon \leq 0.1$, and $0 < \delta < \Delta_M$. In particular, $M \geq 8$ implies $\log M > 0$, so the value $\Delta_M = d_*/(M^2 \log M)$ in (S12.6) is a well-defined positive source cap. Lemma A.36 gives, for every defining pair and event in (S12.27),

$$B_{\mu_{N,M},A}(z, E) \leq e^\varepsilon B_{\mu_{N,M},A}(z', E) + \delta. \quad (\text{S12.29})$$

Because probabilities are nonnegative, $\varepsilon \leq 0.1$, and $\delta < \Delta_M$,

$$\begin{aligned} e^\varepsilon B_{\mu_{N,M},A}(z', E) + \delta & \leq e^{0.1} B_{\mu_{N,M},A}(z', E) + \delta \\ & < e^{0.1} B_{\mu_{N,M},A}(z', E) + \Delta_M. \end{aligned} \quad (\text{S12.30})$$

The weak inequality required in (S12.27) follows. The scalar inequalities from Proposition A.5, (S12.27), and (S12.30), together with the exact total kernel typing, check every component of the source interface separately. No accuracy, realizability, or properness property has been added to that learner set, and no private eligibility is inferred without the event-level derivation (S12.29)–(S12.30). This proves (S12.28). $\square$

The fixed-seed adjacency lemma, the single invocation of Assumption 2, and the common seed mixture prove (S12.4). Proposition A.16 then verifies the exact source-cap typing and parameters in (S12.6).

A.13 Prior-averaged product risk

Set $\eta = 2^{-8}$ and $B = B_{\mu_{N,M},A}$. In the ideal experiment let

$$\Xi \sim \mu_{N,M}^{\otimes k}, \quad S^{\text{id}} \mid \Xi \sim (P_Q^{c_T})^n, \quad H^{\text{id}} \sim A(S^{\text{id}}). \quad (\text{S13.3})$$

The conclusion sought is

$$\mathbb{E} R_{P_Q}(H^{\text{id}}, c_T) > 2^{-9}. \quad (\text{S13.4})$$

Proposition A.17 (Hard-prior loss of the certified simulator). *Under Proposition A.4 and Proposition A.16, if the total simulator $B = B_{\mu_{N,M},A}$ is instantiated at their identical admissible N, M and public prior $\mu_{N,M}$, then*

$$\mathbb{E}_{\substack{(T,Q) \sim \mu_{N,M} \\ Z \sim (Q^{\tau_T})^M \\ B}} R_Q(B(Z), \tau_T) > \eta = 2^{-8}. \quad (\text{S13.9})$$

The expectation includes the prior draw, exactly M i.i.d. realizable one-block input rows conditional on that draw, all simulator preprocessing, and any internal coins of A used by the simulator.

Proof. Proposition A.16 gives every membership clause needed by the universal learner quantifier in Proposition A.4: B is total on the entire arbitrary labeled input space of exact size M , its output is an arbitrary member of $\{0, 1\}^{[N]}$, and it satisfies every event-level one-row replacement inequality at privacy parameters

$$\left(0.1, \frac{d_*}{M^2 \log M}\right). \quad (\text{S13.10})$$

Both propositions use the same finite ordered domain $[N]$, thresholds τ_t , population-risk convention, source constants, and integers N, M . Thus no convention, object, sample-size, or privacy bridge remains to prove.

The prior in Proposition A.4 was selected before the universal learner quantifier. Although the code of B uses the public $\mu_{N,M}$, it is therefore one of the prior-aware kernels already covered by that quantifier. Substituting this exact B into Proposition A.4 yields (S13.9) directly. The substitution neither conditions on overflow nor changes the simulator on any input, so it introduces no residual. $\square$

Proposition A.18 (Prior-averaged product-risk transfer). *Under Propositions A.9, A.10, A.14, and A.17, the ideal experiment (S13.3) obeys the exact product-risk lower bound (S13.4).*

Proof. Define only for this proof

$$\mathcal{L}_{\text{act}} := \mathbb{E}_{\substack{(T,Q) \sim \mu_{N,M} \\ Z \sim (Q^{\tau_T})^M \\ B}} R_Q(B(Z), \tau_T) \quad (\text{S13.11})$$

and

$$\mathcal{L}_{\text{id}} := \mathbb{E}_{\text{ideal}} R_{Q_J}(D_J H^{\text{id}}, \tau_{T_J}). \quad (\text{S13.12})$$

These are proof-local names for the two exact quantities already typed by the dependencies. Proposition A.17 gives

$$\mathcal{L}_{\text{act}} > 2^{-8}. \quad (\text{S13.13})$$

Proposition A.14 and the notation $p_{\text{ov}} = \Pr(U > M)$, with its bound supplied by Proposition A.10, give

$$\mathcal{L}_{\text{id}} \geq \mathcal{L}_{\text{act}} - p_{\text{ov}}. \quad (\text{S13.14})$$

Combining (S13.13) with the weak inequality (S13.14) preserves strictness:

$$\mathcal{L}_{\text{id}} > 2^{-8} - p_{\text{ov}}. \quad (\text{S13.15})$$

Proposition A.10 gives the strict upper bound $p_{\text{ov}} < 2^{-9}$. Since subtraction reverses the comparison in the subtracted term,

$$2^{-8} - p_{\text{ov}} > 2^{-8} - 2^{-9} = 2 \cdot 2^{-9} - 2^{-9} = 2^{-9}. \quad (\text{S13.16})$$

Equations (S13.15)–(S13.16) therefore yield

$$\mathcal{L}_{\text{id}} > 2^{-9}. \quad (\text{S13.17})$$

The overflow event has been charged exactly once: it occurs only in (S13.14), and no later equality or bound adds another copy.

Finally, Proposition A.9 identifies (S13.12), with no residual, as

$$\mathcal{L}_{\text{id}} = \mathbb{E}_{\substack{\Xi \sim \mu_{N,M}^{\otimes k} \\ S^{\text{id}} \sim (P_{\mathbf{Q}}^{cT})^n \\ H^{\text{id}} \sim A(S^{\text{id}})}} R_{P_{\mathbf{Q}}}(H^{\text{id}}, c_T). \quad (\text{S13.18})$$

Substituting (S13.18) into (S13.17) proves (S13.4). The equality in (S13.18) is taken under the unconditional ideal experiment; the conditional i.i.d. sample law is invoked only after fixing Ξ , and no conditioning on $U \leq M$ appears. $\square$

Proposition A.17 applies the public prior only after source-cap membership is established. The ledger in Proposition A.18 preserves strictness and proves (S13.4).

A.14 A fixed hard product instance

For $\mathbf{z} = ((t_i, Q_i))_{i=1}^k \in \mathcal{I}_N^k$, define

$$\Phi_A(\mathbf{z}) = \mathbb{E}_{\substack{S \sim (P_{\mathbf{Q}}^{cT})^n \\ \rho_A}} R_{P_{\mathbf{Q}}}(A_{\rho_A}(S), c_t). \quad (\text{S14.3})$$

Proposition A.19 (Fixed support point from a strict prior average). *Under Proposition A.18, fix its same k, N, M, n, A , finite prior $\mu_{N,M}$, and ideal experiment. Then there exists a deterministic vector*

$$\mathbf{z}^* \in \text{supp}(\mu_{N,M}^{\otimes k}) \subseteq \mathcal{I}_N^k \quad (\text{S14.8})$$

such that

$$\mathbb{E}_{\substack{S \sim (P_{\mathbf{Q}^*}^{cT})^n \\ \rho_A}} R_{P_{\mathbf{Q}^*}}(A_{\rho_A}(S), c_{t^*}) > 2^{-9}. \quad (\text{S14.9})$$

Once $\mathbf{z}^*$ is selected, the only randomness in (S14.9) is the fixed-instance sample and the learner coins.

Proof. Let

$$\mathcal{S} = \text{supp}(\mu_{N,M}^{\otimes k}). \quad (\text{S14.10})$$

Proposition A.18 supplies a finite probability law, so $\mathcal{S}$ is finite. For each $\mathbf{z} \in \mathcal{S}$, put

$$w_{\mathbf{z}} = \mu_{N,M}^{\otimes k}(\{\mathbf{z}\}). \quad (\text{S14.11})$$

Then $w_{\mathbf{z}} > 0$ and $\sum_{\mathbf{z} \in \mathcal{S}} w_{\mathbf{z}} = 1$.

Conditional on $\Xi = \mathbf{z}$, the sample and output experiment in Proposition A.18 is exactly the fixed-instance experiment defining $\Phi_A(\mathbf{z})$. Finite expansion of the outer prior expectation therefore gives

$$\begin{aligned} \mathbb{E}_{\substack{\Xi \sim \mu_{N,M}^{\otimes k} \\ S^{\text{id}} \sim (P_Q^{\text{id}})^n \\ H^{\text{id}} \sim A(S^{\text{id}})}} R_{P_Q}(H^{\text{id}}, c_T) \\ = \sum_{\mathbf{z} \in \mathcal{S}} w_{\mathbf{z}} \Phi_A(\mathbf{z}). \end{aligned} \quad (\text{S14.12})$$

This is finite conditional expectation. It does not assert that the unconditional prior mixture is iid from one deterministic distribution.

The left side of (S14.12) is strictly greater than 2^{-9} by Proposition A.18; hence

$$\sum_{\mathbf{z} \in \mathcal{S}} w_{\mathbf{z}} \Phi_A(\mathbf{z}) > 2^{-9}. \quad (\text{S14.13})$$

If every $\mathbf{z} \in \mathcal{S}$ had $\Phi_A(\mathbf{z}) \leq 2^{-9}$, then

$$\sum_{\mathbf{z} \in \mathcal{S}} w_{\mathbf{z}} \Phi_A(\mathbf{z}) \leq \sum_{\mathbf{z} \in \mathcal{S}} w_{\mathbf{z}} 2^{-9} = 2^{-9}, \quad (\text{S14.14})$$

contradicting (S14.13). Thus some $\mathbf{z}^* \in \mathcal{S}$ has $\Phi_A(\mathbf{z}^*) > 2^{-9}$, which is (S14.9). $\square$

Proposition A.20 (Same-instance PAC contradiction). *Under Proposition A.19 and Proposition A.6, fix the deterministic vector $\mathbf{z}^*$ extracted in (S14.8). Then*

$$2^{-9} < \Phi_A(\mathbf{z}^*) \leq 2^{-12}, \quad (\text{S14.15})$$

which is impossible.

Proof. The extraction proposition gives

$$\Phi_A(\mathbf{z}^*) > 2^{-9}. \quad (\text{S14.16})$$

This lower bound concerns the fixed P_{Q^*} , fixed target c_{t^*} , exact sample size n , fixed randomized learner A , its internal coin law, and the setting-defined population 0-1 risk. No component is changed by selecting $\mathbf{z}^*$.

Because $\mathbf{z}^* \in \mathcal{I}_N^k$, Proposition A.6, which is the pointwise consequence of Assumption 3, applies to this same vector:

$$\begin{aligned} \Phi_A(\mathbf{z}^*) &= \mathbb{E}_{S \sim (P_{Q^*}^{c_{t^*}})^n} R_{P_{Q^*}}(A_{\rho_A}(S), c_{t^*}) \\ &\leq \alpha_0 + \beta_0 = 2^{-13} + 2^{-13} = 2^{-12}. \end{aligned} \quad (\text{S14.17})$$

Equations (S14.16) and (S14.17) concern literally the same deterministic number; neither includes a prior average. Finally,

$$2^{-9} = 8 \cdot 2^{-12} > 2^{-12}, \quad (\text{S14.18})$$

so (S14.16)–(S14.17) contradict one another. $\square$

Finite-support extraction produces one deterministic $\mathbf{z}^*$ with $\Phi_A(\mathbf{z}^*) > 2^{-9}$, while the pointwise PAC result gives $\Phi_A(\mathbf{z}^*) \leq 2^{-12}$. Proposition A.20 records this same-instance contradiction.

A.15 Nonasymptotic closure

The constants are those in (S5.C1), fixed before choosing

$$k \geq 2, \quad N \geq N_0, \quad n \geq 1, \quad (\text{S15.T2})$$

the admissible privacy pair, and the arbitrary randomized map

$$A : (X_{k,N} \times \{0, 1\})^n \longrightarrow \mathcal{H}_{k,N}. \quad (\text{S15.T4})$$

Lemma A.37 (Impossibility of the negated candidate bound). *Under Assumptions 1, 2, and 3, Proposition A.5, and Proposition A.20, fix any candidate in the order (S15.T2)–(S15.T4), with the privacy pair chosen between the integer tuple and the learner. If*

$$n < ak \log_2^* N, \quad (\text{S15.1})$$

then the preceding propositions yield a contradiction. Hence no such fixed candidate can satisfy (S15.1).

Proof. The constants in (S5.C1), including $a = b_*/16$, and N_0 were fixed by Proposition A.5 before the present candidate was chosen. Apply that proposition to the fixed candidate and the strict hypothesis (S15.1). It yields, at the exact integer budget

$$M = \max \left\{ 8, \left\lceil \frac{4n}{k} \right\rceil \right\},$$

the strict hard-regime and source-privacy conclusions

$$N \geq N_*, \quad 8 \leq M < b_* \log_2^* N, \quad 0 < \varepsilon \leq 0.1, \quad 0 < \delta < \frac{d_*}{M^2 \log M}. \quad (\text{S15.2})$$

The same proposition retains, rather than replaces, the primitive conjunction

$$0 < \delta \leq \frac{1}{n \log(n+1)}, \quad 0 < \delta \leq \frac{c_\delta}{M^2 \log(M+1)}. \quad (\text{S15.3})$$

The fixed learner still satisfies its exact Assumption 2 and Assumption 3 premises; neither is altered by the scalar specialization (S15.2). Therefore the complete premise package for Proposition A.20 holds at this identical candidate. Applying that proposition, without reopening any of its internal dependencies, gives one fixed product instance (P_{Q^*}, c_{t^*}) for which

$$\mathbb{E}_{S \sim (P_{Q^*}^{c_{t^*}})^n}_{\rho_A} R_{P_{Q^*}}(A_{\rho_A}(S), c_{t^*}) > 2^{-9} \quad (\text{S15.4})$$

and, for literally the same fixed distribution, target, size- n iid sample law, randomized learner, internal-coin law, and population risk,

$$\mathbb{E}_{S \sim (P_{Q^*}^{c_{t^*}})^n}_{\rho_A} R_{P_{Q^*}}(A_{\rho_A}(S), c_{t^*}) \leq 2^{-12}. \quad (\text{S15.5})$$

But

$$2^{-9} = 8 \cdot 2^{-12} > 2^{-12}, \quad (\text{S15.6})$$

so no real number can satisfy both (S15.4) and (S15.5). This contradicts (S15.1).

There is no exceptional boundary branch. Proposition A.5 derives (S15.2)–(S15.3) with the same strict inequalities when $N = N_0$, $M = 8$, $n = 1$, $n < k$, or $k = 2$, and Proposition A.20 applies to that same candidate and exact fixed-instance experiment. No floor, ceiling, endpoint, privacy parameter, or sample size is changed in this invocation. $\square$

Proposition A.21 (Nonasymptotic candidate closure). *Fix the universal constants in (S5.C1) and the fixed N_0 before candidate quantification. Under Assumptions 1, 2, and 3, every candidate quantified in that same candidate order satisfies*

$$n \geq ak \log_2^* N. \quad (\text{S15.7})$$

This is a deterministic pointwise implication for the exact candidate. It makes no assertion outside the approved candidate regime.

Proof. Fix an arbitrary candidate covered by the proposition. If (S15.7) failed, then the total order on the real numbers would give its exact strict negation

$$n < ak \log_2^* N. \quad (\text{S15.8})$$

No integer rounding is inserted into this logical negation. But Lemma A.37 proves that (S15.8) is impossible for a candidate satisfying the three primitive assumptions. Therefore (S15.7) holds. Because the constants were fixed before the arbitrary candidate and the candidate was otherwise arbitrary within that order, the conclusion has the quantifier order stated in the formalized goal. $\square$

Lemma A.37 excludes the exact strict negation for every candidate in this order. Proposition A.21 therefore yields the weak nonasymptotic lower bound with no rounding or scope change.

A.16 Tower-diagonal specialization

For compactness in the following proof, write

$$T_s = \text{Tow}_2(s) \quad (s \geq 1)$$

Lemma A.38 (Exact tower inversion and floor arithmetic). *Under the tower and iterated-logarithm definitions in Section 1, the map $s \mapsto T_s$ is strictly increasing and unbounded, with $T_s \geq 2^s$ for every integer $s \geq 1$. The function $u \mapsto \log_2^* u$ is nondecreasing on $(0, \infty)$. Moreover, for every integer $r \geq 1$,*

$$\log_2^* T_r = r \quad (\text{S16.1})$$

and, for every integer $r \geq 2$,

$$\lfloor \log_2(T_r + 1) \rfloor = T_{r-1}. \quad (\text{S16.2})$$

Proof. Because $T_1 = 2$ and $T_{s+1} = 2^{T_s}$, induction gives $T_s \geq 2^s$. Indeed, the claim is true at $s = 1$, and the elementary induction $2^s \geq s + 1$ gives

$$T_{s+1} = 2^{T_s} \geq 2^{2^s} \geq 2^{s+1}.$$

Also $2^m > m$ for every integer $m \geq 2$, so $T_{s+1} > T_s$. Thus the tower is strictly increasing and, from $T_s \geq 2^s$, unbounded.

For monotonicity of log-star, first note that for $u > 1$,

$$\log_2^* u = 1 + \log_2^*(\log_2 u). \quad (\text{S16.3})$$

If $0 < u \leq v$, induction on $\log_2^* v$, using the monotonicity of $\log_2$ and (S16.3), gives $\log_2^* u \leq \log_2^* v$. The base case $\log_2^* v = 0$ means $v \leq 1$, so also $u \leq 1$, and both log-star values are zero. In the

induction step, $u \leq 1$ is immediate; if $u > 1$, apply the induction hypothesis to $0 < \log_2 u \leq \log_2 v$ in (S16.3).

For $0 \leq j \leq r-1$, repeated use of the tower recursion yields

$$\log_2^{(j)} T_r = T_{r-j}. \quad (\text{S16.4})$$

In particular, the $(r-1)$ -fold iterate equals $T_1 = 2 > 1$, whereas the r -fold iterate equals $\log_2 2 = 1$. By the minimum in the definition of log-star, (S16.1) follows exactly.

Finally, put $q = T_{r-1}$. Then $T_r = 2^q > 1$, and hence

$$2^q < T_r + 1 < 2T_r = 2^{q+1}.$$

Taking base-two logarithms gives

$$q < \log_2(T_r + 1) < q + 1.$$

Since q is an integer, its floor is exactly $q = T_{r-1}$, proving (S16.2). $\square$

Proposition A.22 (Exact diagonal structural identities). *Under the primitive diagonal specialization $r \in \mathbb{Z}_{\geq 2}$, $k = r$, $N = T_r$, Lemma A.1, Lemma A.2, Proposition A.1, and Lemma A.38,*

$$\text{VC}(C_{r,T_r}) = r, \quad \text{LD}(C_{r,T_r}) = rT_{r-1}, \quad |C_{r,T_r}| = (T_r + 1)^r. \quad (\text{S16.5})$$

With the setting convention that unadorned log is natural,

$$rT_{r-1} \log 2 < \log |C_{r,T_r}| < \frac{3}{2} rT_{r-1} \log 2. \quad (\text{S16.6})$$

In particular, $\log |C_{r,T_r}| = \Theta(rT_{r-1})$ with universal hidden constants.

Proof. The structural identities give, after the exact substitution $k = r, N = T_r$,

$$\text{VC}(C_{r,T_r}) = r,$$

$$\text{LD}(C_{r,T_r}) = r \lfloor \log_2(T_r + 1) \rfloor,$$

and $|C_{r,T_r}| = (T_r + 1)^r$. Equation (S16.2) in Lemma A.38 turns the middle display into rT_{r-1} , proving every exact identity in (S16.5).

For (S16.6), write $T_r = 2^{T_{r-1}}$. The same strict comparison used for the floor gives

$$T_{r-1} \log 2 < \log(T_r + 1) < (T_{r-1} + 1) \log 2.$$

Since $r \geq 2$ implies $T_{r-1} \geq T_1 = 2$, $T_{r-1} + 1 \leq (3/2)T_{r-1}$. Multiplication by r and the exact identity $\log |C_{r,T_r}| = r \log(T_r + 1)$ prove (S16.6). The constants $\log 2$ and $(3/2) \log 2$ are absolute and independent of r . $\square$

Lemma A.39 (Iterated logarithm of the diagonal Littlestone dimension). *Under Lemma A.38 and Proposition A.22, for every integer $r \geq 2$,*

$$r - 1 \leq \log_2^* \text{LD}(C_{r,T_r}) \leq r. \quad (\text{S16.7})$$

Consequently,

$$\frac{r}{2} \leq \log_2^* \text{LD}(C_{r,T_r}) \leq r, \quad (\text{S16.8})$$

so $\log_2^ \text{LD}(C_{r,T_r}) = \Theta(r)$ with constants $1/2$ and 1 , independent of r .*

Proof. Let $q = T_{r-1}$. Proposition A.22 gives

$$\text{LD}(C_{r,T_r}) = rq \geq q = T_{r-1}. \quad (\text{S16.9})$$

We next prove the upper comparison $rq \leq T_r = 2^q$. At $r = 2$, $q = T_1 = 2$ and both sides equal 4. For $r \geq 3$, induction gives $T_{r-1} \geq r$: the base is $T_2 = 4 \geq 3$, and $T_s \geq s + 1$ implies $T_{s+1} = 2^{T_s} \geq 2^{s+1} \geq s + 2$. Also, $2^x \geq x^2$ for every integer $x \geq 4$; this follows by induction from equality at $x = 4$ and $2x^2 \geq (x+1)^2$ for $x \geq 3$. Therefore, for $r \geq 3$,

$$T_r = 2^q \geq q^2 \geq rq. \quad (\text{S16.10})$$

Combining (S16.9)–(S16.10), monotonicity of log-star from Lemma A.38, and its exact tower inversion gives

$$\log_2^* T_{r-1} = r - 1 \leq \log_2^*(rq) \leq \log_2^* T_r = r,$$

which is (S16.7). Finally, $r - 1 \geq r/2$ for every $r \geq 2$, yielding (S16.8) and the stated uniform Θ -constants. $\square$

Proposition A.23 (Tower-diagonal Rate Specialization Bridge). *Fix the universal constants*

$$a, c_\delta, \varepsilon_0, \alpha_0, \beta_0, N_0$$

from Proposition A.21, before any candidate is chosen. Thus $a = b_/16$, $c_\delta = d_*$, $\varepsilon_0 = 0.1$, and $\alpha_0 = \beta_0 = 2^{-13}$. Define the finite deterministic index*

$$r_0 = \min\{s \in \mathbb{Z}_{\geq 2} : T_s \geq N_0\}. \quad (\text{S16.11})$$

Under Assumptions 1, 2, and 3, every diagonal candidate with integer $r \geq r_0$, $k = r$, and $N = T_r$ satisfies

$$n \geq ar^2 \quad (\text{S16.12})$$

and

$$n \geq a \text{VC}(C_{r,T_r}) \log_2^* \text{LD}(C_{r,T_r}). \quad (\text{S16.13})$$

Moreover,

$$\frac{1}{2}r^2 \leq \text{VC}(C_{r,T_r}) \log_2^* \text{LD}(C_{r,T_r}) \leq r^2. \quad (\text{S16.14})$$

Thus (S16.12)–(S16.13) are exactly $n = \Omega(r^2) = \Omega(\text{VC} \log_2^ \text{LD})$, with constants independent of r , and only within the approved candidate/privacy/PAC regime.*

Proof. Lemma A.38 proves that T_s is increasing and unbounded, so the set in (S16.11) is nonempty and r_0 is a finite integer depending only on the already-fixed N_0 . For every $r \geq r_0$, one has $r \geq 2$ and $T_r \geq T_{r_0} \geq N_0$. Therefore $k = r, N = T_r$ meet the structural range of Proposition A.21.

No privacy condition is absorbed in this substitution. The candidate has the exact budget

$$m_{n,r} = \max \left\{ 8, \left\lceil \frac{4n}{r} \right\rceil \right\},$$

and Assumption 1 remains exactly

$$n \in \mathbb{Z}_{\geq 1}, \quad 0 < \varepsilon \leq \varepsilon_0, \quad 0 < \delta \leq \min \left\{ \frac{1}{n \log(n+1)}, \frac{c_\delta}{m_{n,r}^2 \log(m_{n,r}+1)} \right\}. \quad (\text{S16.15})$$

The learner remains an arbitrary randomized map

$$A : (X_{r,T_r} \times \{0,1\})^n \longrightarrow \{0,1\}^{X_{r,T_r}}$$

satisfying the exact central replacement-DP and distribution-free realizable-PAC assumptions at this same candidate. Proposition A.21 therefore applies without a mode or scope conversion and gives

$$n \geq ar \log_2^* T_r.$$

Equation (S16.1) in Lemma A.38 makes the right-hand side exactly ar^2 , proving (S16.12). No asymptotic term is dropped in this equality.

By Proposition A.22 and Lemma A.39,

$$\text{VC}(C_{r,T_r}) = r, \quad \frac{r}{2} \leq \log_2^* \text{LD}(C_{r,T_r}) \leq r.$$

Multiplication proves (S16.14). In particular, the upper half of (S16.14) and (S16.12) give

$$n \geq ar^2 \geq a \text{VC}(C_{r,T_r}) \log_2^* \text{LD}(C_{r,T_r}),$$

which is (S16.13) with the same inherited constant a .

The finite initial indices are not hidden inside an r -dependent constant. At $r = 2$, the arithmetic itself is exact:

$$T_1 = 2, \quad T_2 = 4, \quad \log_2^* T_2 = 2, \quad \lfloor \log_2(5) \rfloor = 2,$$

and hence $\text{VC} = 2$, $\text{LD} = 4$, $|C| = 25$, and $\log_2^* \text{LD} = 2$. The same structural identities hold for every $2 \leq r < r_0$. By the minimality of r_0 and monotonicity of the tower, every such index has $T_r < N_0$. Therefore Proposition A.21 is inapplicable, so no learning lower-bound conclusion is asserted for those indices. The asymptotic statements begin at the single fixed index r_0 . $\square$

Proposition A.24 (Diagonal separation and unresolved scales). *Under Proposition A.22 and Lemma A.39, for every integer $r \geq 2$,*

$$2r - 1 \leq \text{VC}(C_{r,T_r}) + \log_2^* \text{LD}(C_{r,T_r}) \leq 2r, \quad (\text{S16.16})$$

and

$$r2^{r-1} \leq \text{VC}(C_{r,T_r})2^{\log_2^* \text{LD}(C_{r,T_r})} \leq r2^r. \quad (\text{S16.17})$$

Consequently, the established lower-bound scale r^2 exceeds the additive expression by an unbounded factor, but

$$r^2 = o\left(\text{VC}(C_{r,T_r})2^{\log_2^* \text{LD}(C_{r,T_r})}\right) \quad (\text{S16.18})$$

and

$$r^2 = o(\log |C_{r,T_r}|). \quad (\text{S16.19})$$

These comparisons concern the size of the proved lower-bound expression; they are not upper bounds on the true sample complexity.

Proof. Substitute $VC = r$ and the two bounds in (S16.7). Addition gives (S16.16), and exponentiation followed by multiplication by r gives (S16.17). In particular,

$$\frac{r^2}{VC(C_{r,T_r}) + \log_2^* LD(C_{r,T_r})} \geq \frac{r}{2} \rightarrow \infty,$$

which proves the unbounded-factor comparison with the additive scale.

For the first remaining scale, the lower half of (S16.17) gives

$$0 \leq \frac{r^2}{VC(C_{r,T_r}) 2^{\log_2^* LD(C_{r,T_r})}} \leq \frac{r}{2^{r-1}}. \quad (\text{S16.20})$$

The right-hand side tends to zero: its ratio at consecutive indices is $(r+1)/(2r) \leq 3/4$ for $r \geq 2$. This proves (S16.18).

For cardinality, (S16.6) and $T_{r-1} \geq 2^{r-1}$ from Lemma A.38 yield

$$0 \leq \frac{r^2}{\log |C_{r,T_r}|} < \frac{r}{T_{r-1} \log 2} \leq \frac{r}{2^{r-1} \log 2} \rightarrow 0,$$

which proves (S16.19). $\square$

Assembly of the tower-diagonal specialization. Lemma A.38 uses the exact convention

$$T_1 = 2, \quad T_{s+1} = 2^{T_s},$$

and the minimum-based definition of log-star to prove

$$\log_2^* T_r = r, \quad \lfloor \log_2(T_r + 1) \rfloor = T_{r-1}.$$

Combining the floor identity with Lemmas A.1 and A.2 and Proposition A.1, Proposition A.22 gives the exact formulas

$$VC(C_{r,T_r}) = r, \quad LD(C_{r,T_r}) = rT_{r-1}, \quad |C_{r,T_r}| = (T_r + 1)^r.$$

Lemma A.39 sharpens the asymptotic notation to the two-sided finite-index inequality

$$r - 1 \leq \log_2^* LD(C_{r,T_r}) \leq r,$$

and hence to

$$\frac{1}{2}r^2 \leq VC(C_{r,T_r}) \log_2^* LD(C_{r,T_r}) \leq r^2.$$

Let r_0 be the fixed index in Proposition A.23. For every $r \geq r_0$, that proposition verifies the unchanged candidate, privacy, and PAC conditions before applying Proposition A.21; it therefore gives, with the same universal a ,

$$n \geq ar^2 \geq a VC(C_{r,T_r}) \log_2^* LD(C_{r,T_r}).$$

All structural identities remain valid for the finite indices $2 \leq r < r_0$, while no learning conclusion is asserted there because $T_r < N_0$.

Finally, Proposition A.24 proves

$$2r - 1 \leq VC(C_{r,T_r}) + \log_2^* LD(C_{r,T_r}) \leq 2r,$$

so the quadratic lower bound exceeds the additive expression by an unbounded factor. The same proposition proves

$$r^2 = o\left(VC(C_{r,T_r}) 2^{\log_2^* LD(C_{r,T_r})}\right), \quad r^2 = o(\log |C_{r,T_r}|).$$

Thus the result remains a partial resolution: it proves neither the stronger exponential milestone nor the $\Omega(\log |C|)$ lower bound, and it supplies neither a universal upper bound nor a matching combinatorial characterization for all finite-Littlestone classes. $\square$

A.17 Proof of the Main Theorem

Proof of Theorem 3.1. Choose b_*, d_*, N_* from Proposition A.2, and then choose $a, c_\delta, \varepsilon_0, \alpha_0, \beta_0, N_0$ as in Lemma A.16 and (S5.C1). These constants are universal and precede every candidate quantifier.

Fix an arbitrary candidate satisfying Assumptions 1, 2, and 3. Proposition A.21 gives

$$n \geq ak \log_2^* N,$$

which proves (1) with the exact fixed-size, replacement-private, population-risk, and unrestricted-improper scope.

For $T_s = \text{ToW}_2(s)$, Lemma A.38 proves $\log_2^* T_r = r$ and the exact floor identity. Proposition A.22 and Lemma A.39 then give (2)–(4). The tower is increasing and unbounded, so the displayed r_0 is finite. For every $r \geq r_0$, Proposition A.23 applies the nonasymptotic bound to the same admissible diagonal candidate and proves (5). Finally, Proposition A.24 gives (6)–(7). This proves all parts of the theorem. $\square$